

Forecasting Intraday Realized Variance with a Dense but Weak Signal: Kernel Backbone, Attribution, Model Class, and Algorithm Design

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Abstract

We study one-step-ahead forecasting of 30-minute realized variance of the S&P 500 over 2005–2024 on a panel of 218,909 bars with 41 exogenous channels, and report a result about data representation rather than about model class. All comparisons are made bar-for-bar on identical rows with Diebold–Mariano inference throughout. The dominant effect we measure is not a modelling choice: one exogenous feed stops publishing on 2023-08-31 while the panel runs eight months past it, and an unlimited forward fill—combined with an availability indicator computed *after* that fill rather than before—presented eight months of a dead feed as live data. Because features are standardised in rolling interquartile units and a frozen series has no dispersion, the resulting division produced feature values at 2,260 rolling IQRs. On the affected window this costs QLIKE 6.37170 against 0.12391 once repaired, a factor of fifty, while every genuine modelling effect in this study lies between 0.0005 and 0.004. Three further feeds terminate the same way inside the evaluation region and were not previously identified. The defect is invisible to ordinary practice—no test fails, no value is missing, the design matrix is dense and finite throughout—and is recoverable from a single comparison we recommend as standard: the mean of each channel’s availability indicator must equal the fraction of rows on which the raw feed actually printed. On the uncorrected panel that check fails for 31 of 41 channels. The repair is an interaction rather than a fix: recording availability honestly changes no feature value, estimating location and scale from observed rows alone leaves the worst channel untouched, and substituting a neutral median without masked estimation is actively harmful, taking a densely-observed channel from 17.6 to 932.6 rolling IQRs; composed, they reach 6.7 from 2,260.4. Three separate attempts at this repair failed before we ran the two-by-two, each because it shipped one half. We then re-run the study’s model comparisons on the corrected panel under a design pre-registered before the results were seen, and report the outcomes against the hypotheses as stated. The winner depends on the data version: on the uncorrected panel a stripped-down 53-column design attains 0.13248 and plain ridge on the 504-feature design attains 0.13617—worse than the 12-column backbone it exists to improve—while on the corrected panel the ordering reverses to

0.13272 and 0.13146. Our own pre-registered falsification condition for the stripped-down design is met at -0.00126 ($t = -5.91$), and we withdraw it: it had won by diluting the corrupted channel twelve-fold, and the raw parameterisation had lost by carrying it faithfully. Two further results survive the correction and one reverses under it. Sweeping rank on the lag axis under two independent nested orthonormal bases—a power-law family and log-polynomials in log-rung—shows the two agreeing from $r = 4$ onward and flat thereafter, spanning 0.00045 over $r \in [4, 12]$, so rank beyond four buys nothing and $r = 8$ at 0.13145 is indistinguishable from all 504 columns at 0.13146. Below $r = 4$ the bases disagree—the power family at $r = 3$ reaches 0.13117 in 135 columns while log-polynomials at the same rank reach 0.13246—so what compresses this design is a particular family of shapes and not a particular dimension, a distinction a single-basis sweep would have reported as an intrinsic rank of three. The 0.00029 by which that 135-column design appears to beat all 504 columns does not survive scrutiny: its Diebold–Mariano statistic runs from -3.96 to -1.30 as the HAC bandwidth is widened, a stationary bootstrap admits zero at block lengths of 1,000 and above, it favours the compressed design in five of eight eras (sign test $p = 0.73$), and deleting one era collapses it to -0.000027 . We therefore report the two as tied, which is the more useful result: a 135-column design and a 504-column one forecast identically, and per-bar the top 1% of observations carry 459% of a differential that nets to nothing; a random-feature approximation to an RBF kernel on the amplitude coordinates gains only 0.00098 against a tree-over-linear premium of 0.00184 to 0.00358, so whatever the trees exploit is not a smooth function of those coordinates. In a k -nearest-neighbour retrieval task, a supervised six-dimensional projection onto the exogenous operator’s singular triplets *beats* the full 516-dimensional ambient view by 0.00540 ($t = -8.9$) on corrected data, having been an exact null on corrupted data, while unsupervised compressions of the same dimension lose by $+0.00450$ to $+0.00552$. One pre-registered hypothesis fails, and we report it as such: pooled over all designs the data-version main effect is 0.00067 against a design main effect of 0.00311, so the claim that data repair outweighs any modelling choice holds on the affected window and in the failing era but not in the twenty-year aggregate. A replication campaign on a second construction—a frozen state-space basis and its lead–lag dynamics, reported separately, on its own evaluation convention, and never merged into the primary tables—re-derives the study’s central laws with different machinery and adds two: the basis’s one-bar lead–lag field is finite-rank, flat on a width plateau of 30–80 directions with cliffs on both sides, and stable under split-half replication at $+0.84$; and the forecast-error second moment’s regime change manifests not as decay of predictability (its own-means R^2 posts its best year in 2020) but as collapse of *complexity*—every dictionary expansion beyond trailing means dies after 2022 except a calendar fragment whose mechanism cannot rot. That campaign retired sixteen further pre-registered arms at their gates and closed with a seven-for-seven family-wise sweep, taking the one-bar deliverable from QLIKE 0.12579 to 0.12299. Finally, we publish a ledger of fifteen claims we made and withdrew during this study, together with the four causes they trace to: one-factor-at-a-time ablation, unvalidated proxy metrics, selection touching the reported loss, and numbers quoted across incompatible data versions. Three of those four are invisible in a finished paper. Our own answers to the corresponding questions were wrong, and we had the code.

1 Introduction

Realized-volatility forecasting sits in an unusual corner of the prediction landscape. The target is among the most forecastable objects in finance—simple heterogeneous-autoregressive (HAR) models on lagged realized variance explain the large majority of out-of-sample variation [Corsi, 2009, Andersen et al., 2003]—yet the increments of skill beyond that autoregressive core are small, fragile, and easy to manufacture spuriously. Machine-learning comparisons in this literature routinely disagree about whether nonlinear models help at all [Christensen et al., 2023, Gu et al., 2020], and we will show that part of that disagreement is mechanical: on our own panel, a hyperparameter search scored on the evaluation set produced an apparent model-class reversal roughly an order of magnitude larger than any effect we can defend under a causal protocol (Section 8.4).

This paper is an attempt to make those increments of skill legible. We work on a single fixed panel—218,934 thirty-minute bars of S&P 500 realized variance spanning 2005–2024, with exogenous predictors organized into eight economic buckets—and we hold one benchmark fixed throughout: an ordinary-least-squares HAR model with calendar features, refit every bar, with out-of-sample QLIKE of 0.13415. Every model is evaluated on the identical row index, so every comparison is bar-for-bar, and every claim of improvement carries a Diebold–Mariano statistic with heteroskedasticity-and-autocorrelation-consistent inference [Diebold and Mariano, 1995, Harvey et al., 1997].

The instrument comes first. Section 6 argues that the target’s own structure determines the forecasting instrument before any exogenous question can be posed, and determines it as a truncated power-law convolution kernel with a session clock. The argument is elementary: the fitted HAR coefficients on rolling means imply a per-lag effective kernel $\varphi(u) = \sum_{k:r_k \geq u} w_k/r_k$, a step function whose breakpoints are the lag rungs. Under this reading HAR is a quadrature rule on a scale-free grid rather than a cascade of economically distinct horizons, and the reading makes predictions the cascade reading does not. Refining the grid should pay: it does, monotonically across six geometric bases, with base-2 beating base-5 by 0.00082 at DM $t = -10.6$, and the conventional daily/weekly/monthly structure faring worse than either. A power-law kernel should beat an exponential one: it does, by 0.011 in forecast loss and by R^2 0.80 against 0.44 in kernel shape. The reading also makes the correspondence to self-exciting point processes [Hawkes, 1971, Bacry et al., 2015b] explicit rather than metaphorical, which means it can be tested—and Section 6.5 tests it. The result splits cleanly. The kernel’s shape behaves: estimated across 41 channels where the log-log slope is scale-invariant, the decay exponent is 1.033 ± 0.036 , agreeing with the Hawkes literature and rejecting both the 1.36 earlier drafts quoted from a plotting constant and the ≈ 1.6 a rough-volatility correspondence implies. The kernel’s sign structure appeared not to: on the estimate this project had been reading, 8 of 12 lags are negative and the integrated kernel is -0.363 , outside the $[0, 1]$ a linear self-exciting model admits. Rebuilding the feature cache reproduces that estimate exactly and then explains it. The archived probe conditions the self-kernel on 41 exogenous channels, several of which are bipower variation, absolute returns and semivariance—volatility measures, and therefore near-substitutes for lagged RV . They absorb the target’s own persistence, and what remains is a contrast between correlated proxies rather than a kernel. The *marginal* self-kernel on identical rows has integrated mass

0.913, no negative mass, and a clean power-law decay. The methodological point generalises beyond this paper: a self-kernel estimated alongside volatility proxies is not a self-kernel, and nothing in the fit signals the problem.

This ordering has an uncomfortable consequence that we state up front rather than defer to a limitations section. Our shared benchmark uses the coarser base-5 grid, so every increment reported downstream includes 0.00082 of backbone misspecification as though it were exogenous or nonlinear content. Section 7.1 accounts for the debt table by table. It changes no ordering, but it dissolves the smallest effects and removes 28% of the headline nonlinearity premium.

Where the information lives. Section 7 measures attribution inside a single model class so that information is never confounded with capacity. Every bucket improves on the backbone, and every improvement is individually decisive, but the gains overlap almost completely: summed within class, the seven singleton buckets give 2.08 times the joint gain. A Frisch–Waugh–Lovell block decomposition attributes 96% of explainable variance to the backbone, and a single mechanism-targeted engineered feature outperforms the 359-column raw exogenous block by a wide margin per column—a comparison we report with its arithmetic and its two caveats, because per-feature normalization is easy to abuse.

Whether extraction requires nonlinearity. Section 8 crosses five estimators with nine feature sets, all causally tuned, all on the shared panel. Tree ensembles hold a modest edge on every feature set, from 0.00184 to 0.00358, and Section 11.2 tests those differences directly: because the shared incumbent cancels from any pairwise differential, a bound that assumes nothing about the dependence between arms resolves all nine comparisons in the tree’s favour after Holm adjustment. The pure-nonlinearity premium on backbone features is 0.00215 against a correctly specified linear ladder. The edge is not a tuning story from any angle we measured: causally tuned linear arms are worse than the untuned incumbent, and a test-set oracle over penalties is worth at most 0.00015. What the trees exploit saturates at additive-plus-pairwise structure anchored to the session clock—which is what a quadratic self-exciting feedback [Blanc et al., 2017] predicts, converting a puzzling ceiling into an expected one.

What algorithm should extract it, and how little of it is needed. Section 9 freezes the feature set and treats estimation as the design space: exact rank-1 rolling least squares, a recursive elastic net via online homotopy [Garrigues and El Ghaoui, 2008], a causal rolling hyperparameter-selection protocol, and a spectral k -nearest-neighbors estimator that embeds trailing residual trajectories with Laplacian eigenmaps [Belkin and Niyogi, 2003]. The section’s contribution is its ablations, and they generalise past the estimator they were built to test: the manifold-embedding step hurts on all eight matched configurations, and a matched retrieval-metric experiment shows the cause is not the out-of-sample extension the design was blamed on—a principal-component projection at the same dimension has no extension step at all and loses nearly as much. On the uncorrected panel every estimated geometry we tried lost to a fixed one; on the corrected panel that statement holds only for the *unsupervised* ones. A supervised six-dimensional projection onto the exogenous operator’s singular

triplets beats the full 516-dimensional ambient view by 0.00540 ($t = -8.9$), having been an exact null before the repair, while principal components and Laplacian eigenmaps at the same dimension still lose by +0.00552 and +0.00450. A distance weights every coordinate by its scale, so a channel standardised by a collapsed denominator dominates the metric outright—which is why this particular result could not have been read correctly on the panel we had.

Section 4 interrupts that thread, because pursuing it further on the panel we had would have been pursuing an artefact. One exogenous feed stops publishing on 2023-08-31 while the panel runs eight months past it; an unlimited forward fill carried its final value across those months, and an availability indicator computed *after* the fill labelled every one of those bars observed. Features are standardised in rolling interquartile units, a frozen series has no dispersion, and the division that standardises it is then a division by nearly nothing: the affected columns reach 2,260 rolling IQRs. The window cost is QLIKE 6.37170 against 0.12391 repaired. Three further feeds terminate the same way inside the evaluation region, and we did not find them until we built the check. That check is one line—each channel’s availability indicator mean must equal its raw print rate—and on the uncorrected panel it fails for 31 of 41 channels. The repair is an interaction: the honest indicator changes no feature value, masked scale estimation alone leaves the worst channel untouched, and a neutral median without masked estimation is actively harmful; composed they reach 6.7 from 2,260.4.

The consequence for the thread above is direct, and it cost us a draft. On the uncorrected panel a stripped-down design—contract each channel’s rung ladder against a fixed u^{-1} weight, run plain ridge on the resulting 41 numbers—reaches QLIKE 0.13248 in 53 columns while plain ridge on the 504-feature design reaches 0.13617, worse than the 12-column backbone it exists to improve. We had made that the paper’s central result. On the corrected panel the ordering reverses, to 0.13272 and 0.13146. The stripped-down design had won by attenuating the corrupted channel twelve-fold and the raw parameterisation had lost by carrying it faithfully; neither number was about parameterisation at all. We re-ran the comparison under a design pre-registered before the results were seen, our own falsification condition for the stripped-down design was met at -0.00126 ($t = -5.91$), and we withdraw it. What survives the correction is narrower and better supported, and narrower again once controlled properly. Sweeping rank on the lag axis under two independent nested orthonormal bases, the two agree from $r = 4$ upward and the curve is flat there—0.00045 across $r \in [4, 12]$ —so rank beyond four buys nothing and $r = 8$ in 340 columns matches all 504. The bases disagree below $r = 4$: an orthonormalised power-law triple reaches 0.13117 in 135 columns while log-polynomials at the same rank reach 0.13246, and the latter’s validation-selected arm loses to the full design outright by 0.00097 ($t = +8.10$). Low-rank compression here is a claim about which shapes, not about how many; a single-basis sweep would have reported an intrinsic dimension of three, and we ran the second basis precisely because this project has already published four claims that a control would have caught. It also would have reported the 135-column design as beating all 504 by 0.00029 at $t = -3.00$, and Section 11 shows why that is not reportable either: the statistic falls to -1.30 under a wider HAC bandwidth, a block bootstrap admits zero, five of eight eras favour it, and deleting one era leaves -0.000027 . The two designs tie. That is the more useful finding, and it is the paper’s thesis applied to itself—a nearly fourfold difference in width buys nothing, while a

dead data feed cost a factor of fifty.

What dominates, and where it does not. Section 10 reports the effects that exceed every modelling choice in this paper. A fixed training window rather than a rolling one is worth 0.11 QLIKE. The imputation defect above is worth a factor of fifty on its window and moves the failing era from $+0.03321$ to -0.00332 . Selecting a shape parameter against the reported loss instead of a validation tail is worth 0.0033. Against these, the model-class effects this literature argues about are 0.002 to 0.004. We pre-registered the stronger version of that claim—that the data-version main effect would exceed the design main effect—and it fails: pooled over six designs the cache effect is 0.00067 against 0.00311. The honest statement is therefore conditional rather than universal. Data representation dominates where the defect binds, catastrophically; averaged over twenty years and over designs that mostly do not touch the affected columns, it does not. We report the failed hypothesis because the conditional version is the one that transfers, and because a pre-registration that only ever confirms is not doing any work.

An independent replication. Section 12 reports a campaign run after the main results froze, on a second construction the primary tables never touch: the base features projected onto a permanently frozen first-window eigenbasis, and the dynamics — lead-lag structure, rotational geometry, error second moments — of the trajectory on that basis. The section exists for the same reason the second orthonormal basis above does: this project no longer trusts a law demonstrated once. Three of the paper’s central claims re-derive there with different machinery (density with uniform generous shrinkage beats selection; the exploitable content is fast and freshness-loving; estimated geometry loses to fixed geometry without supervision), and the campaign contributes two laws the primary basis could not see — a finite-rank lead-lag field flat on a wide plateau of basis truncations, and a regime change that destroys a channel’s *complexity* while strengthening its first-order structure. Every arm was pre-registered with a recorded expectation; sixteen died at their gates; every survivor passed a Romano–Wolf sweep within its claim family, the closing family of seven with no casualties.

Commitments. Three run through the paper. First, one panel, one benchmark: no result is quoted across incompatible evaluation regimes, and superseded experiments contribute no number. Second, significance before narrative: differences without a DM statistic are labeled unresolved, quantities we have not computed are marked in the text rather than elided, and every multiple-comparison family is adjusted (Section 11)—which cost us three inherited claims and one that had never been computed. Third, negative results are reported at the same altitude as positive ones—the failures of PCA compression, deep sequence models, hyperparameter tuning, and our own manifold embedding are load-bearing facts about the dense-but-weak regime, not footnotes. Section 14 collects the threats that could change a conclusion, of which the largest are the benchmark debt above, a target winsorization that clips 10% of observations by construction, and a diurnal adjustment that is itself a forecast. To those we add a fourth commitment adopted partway through this study and applied retroactively: Section 13 publishes the fifteen claims we made and withdrew, and the four

causes they trace to. We think the ledger is the most useful thing in the paper for a reader deciding how much of the rest to believe.

2 Related Work

Realized volatility and the HAR class. Realized variance as a nonparametric estimator of integrated variance from high-frequency returns is due to Andersen et al. [2003], and the heterogeneous autoregressive model of Corsi [2009] remains the reference specification: a regression of future realized variance on daily, weekly and monthly trailing averages, motivated as a cascade of market participants operating at different horizons and as a parsimonious approximation to long memory. Extensions add jump components [Andersen et al., 2007, Barndorff-Nielsen and Shephard, 2004], correct for measurement error in the realized proxy [Bollerslev et al., 2016], or couple returns and realized measures in a single likelihood [Hansen et al., 2012].

Our reading of HAR in Section 6.2 differs from the cascade interpretation in a way that is empirically consequential. We treat the lag ladder not as a set of economically distinct horizons but as a quadrature grid for a single continuous decay kernel, which predicts that refining the grid should pay monotonically. Table 3 confirms this on our panel, and the conventional daily/weekly/monthly structure is *worse* than a denser geometric ladder. To our knowledge the lag grid has not previously been treated as a tunable quadrature in this literature.

Hawkes processes and endogenous volatility. Self-exciting point processes [Hawkes, 1971, Hawkes and Oakes, 1974] model activity as a superposition of exogenous arrivals and endogenously triggered offspring, with the branching ratio $n = \int \varphi$ measuring the endogenous share. Applications to financial data are surveyed by Bacry et al. [2015b]. The empirical kernel is consistently found to be power-law, and estimates of n close to unity have been read as evidence that markets operate near criticality [Hardiman et al., 2013]; Filimonov and Sornette [2012] argue that such estimates are inflated when the background intensity is time-varying and unmodelled, which is directly relevant to intraday data with strong periodicity [Andersen and Bollerslev, 1997].

Two extensions bear on our results. Nonlinear Hawkes processes admit inhibition [Brémaud and Massoulié, 1996], which is the only class in which a kernel with negative lobes—such as the one we estimate—is well defined. Quadratic Hawkes [Blanc et al., 2017] makes the feedback a quadratic form in the past, generating clustering, fat tails and the Zumbach effect [Zumbach, 2009] endogenously; it predicts extractable structure saturating at pairwise order, which is what Section 8.7 measures. For the multivariate case, estimation of low-rank and sparse kernel matrices is treated by Bacry et al. [2015a] and Achab et al. [2017].

Rough volatility. Jaisson and Rosenbaum [2016] show that nearly unstable Hawkes processes with heavy-tailed kernels converge under rescaling to rough fractional processes, supplying a microstructural foundation for the rough-volatility models of Gatheral et al. [2018] and El Euch and Rosenbaum [2019]; see also Bennedsen et al. [2022] on separating short- and long-horizon behaviour. This literature is the closest competitor to our backbone, since

it too predicts a power-law prediction kernel, and Section 6.8 notes that our battery does not yet contain a fractional-kernel arm. We regard that as the most consequential missing benchmark in the paper.

Machine learning for volatility forecasting. Gu et al. [2020] establish the modern protocol for large-scale model comparison in asset pricing, and Christensen et al. [2023] apply it to realized volatility, reporting gains for nonlinear methods. Results in this area disagree about whether nonlinearity helps, and by how much. Our Section 8.4 contributes a measured account of one mechanism behind that disagreement: a hyperparameter search scored on the evaluation panel produced, on our own data, an apparent model-class reversal roughly an order of magnitude larger than any effect we can defend under a causal protocol. We therefore read cross-study disagreement as partly a protocol artifact rather than as substantive disagreement about volatility.

Our finding that dense shrinkage dominates sparse selection is consistent with the weak-and-dense-signal view of return predictability, and our attribution machinery follows the Frisch–Waugh–Lovell tradition [Frisch and Waugh, 1933, Lovell, 1963]. For inference we use Diebold–Mariano tests [Diebold and Mariano, 1995] with the small-sample correction of Harvey et al. [1997], the model confidence set of Hansen et al. [2011], and QLIKE as the primary loss on the robustness grounds of Patton [2011]. Section 8.6 reports a property of that loss on our panel—that it systematically rewards attenuated forecasts—which we have not seen quantified elsewhere in this setting.

3 Data

Our dataset comprises 30-minute intraday observations of the S&P 500 beginning in January 2005. The forecasting target is realized *variance*, measured as the sum of squared 5-minute log-returns within each 30-minute bar ($RV_t = \text{sumret2}$; see Equation (1)). We use “realized variance” throughout for the target itself and reserve “volatility” for its square root, on which the models are estimated (Section 5.2.2).

3.1 Exogenous Variables

In addition to the target series, we incorporate variables spanning several information categories:

- **Higher-order return moments:** bipower variation, realized semi-variance, autocovariance, absolute returns, cubic returns, and quartic returns.
- **Liquidity and microstructure:** trading volume, turnover (total, buy-initiated, sell-initiated), effective spread, quoted spread, and number of trades.
- **Cross-sectional stock factors:** equal-weighted and value-weighted aggregates of stock-level return moments, bipower variation, semi-variance, and absolute returns.
- **S&P 500 and sentiment:** SPY trading activity (turnover, buy/sell turnover) and StockTwits social media metrics (attention, sentiment score, sentiment count).

- **Implied volatility:** VIX, VVIX, and VIX3M.
- **Volatility demand:** Options order flow indicators (SPX open/close, all options open/close, open only).

3.2 Feature Accounting

Because several claims in this paper are per-column (Section 7.3) and because the design width is itself part of the dense-but-weak argument, the mapping from raw series to design columns must be explicit rather than implied. Three counts circulate in this project—41 exogenous channels, a 359-column exogenous block, and a 529-column design matrix—and they cannot all be reconciled by the stated expansion rule of $|\mathcal{L}| = 6$ rolling means per channel, since $41 \times 6 = 246$. **[pending: a per-bucket table giving channels, the lag set applied to each, derived columns, and the reconciliation to 529; no per-column claim in this paper should be read as final until this table exists, and the ratio in Equation (24) depends on it directly]**

3.3 Data Cleaning

The raw data is gridded to a regular 30-minute frequency and filtered to exclude non-trading periods: Friday evenings after 20:00, all of Saturday, and Sunday mornings before 18:30. Overnight gaps in stock factors and volatility demand measures are forward-filled with neutral values. After cleaning and lag feature generation—which requires a burn-in of 3,125 bars for the longest lag on the base-5 ladder—the frozen evaluation panel comprises exactly 218,934 observations, and every arm reported in this paper is scored on those identical rows.

3.4 A Defect in That Fill

The sentence above—“forward-filled with neutral values”—understates what the pipeline did, and the understatement was expensive enough to report as a finding rather than a correction. Section 4 is that report. It is placed immediately after this section because no result in the rest of the paper is interpretable without it: the panel described above is not the panel most of our earlier drafts were fitted on.

4 Stale Data as the Dominant Effect

This section reports the largest effect in the paper. It is not a modelling effect. A single exogenous feed stopped publishing partway through the sample, the pipeline did not notice, and the resulting artefact was worth more than every model-class, basis, penalty and shrinkage decision reported elsewhere in this study combined—by three orders of magnitude on the affected window and by roughly a factor of ten pooled over twenty years.

We report it as a finding rather than as a correction because the failure is invisible to ordinary practice. No test fails. No value is missing. Nothing is infinite or negative. The pipeline produces a dense, finite, plausibly-scaled design matrix throughout, and the defect

Table 1: Raw print rate against implied session width, by channel group. A channel that publishes only inside its own session is correctly unavailable for most of a near-24-hour grid; the two columns agree throughout, so the panel’s sparsity is a calendar, not a defect.

group	n	print rate	session
index bar quantities	9	0.980	24h
equity (*_ewstock, *_vwstock)	22	0.623	04:30–20:00 (0.667)
social media	3	0.66–0.75	24h, later start
vix, vix3m, vvix	3	0.16–0.35	10:00–16:30 (0.292)
volatility demand	4	0.220	10:00–19:00 (0.354)

is detectable only by comparing the pipeline’s own bookkeeping against the raw inputs it came from—a comparison that, so far as we can determine, is not standard.

4.1 What Happened

The panel is a 30-minute grid spanning 2005-01-02 to 2024-04-30, covering Sunday 18:30 through Friday 20:00—approximately 24 hours a day, five days a week. The 41 exogenous channels are not all published on that grid. Most are US cash- or extended-session equity quantities, and each prints only inside its own session. Table 1 gives the measured print rates against the session width each channel’s own publication times imply.

Averaged across the 41 channels, 0.638 of panel rows carry a genuine print. The remaining 0.362 must be filled, and the pipeline filled them by carrying the last observed value forward without limit.

The volatility-demand family stops publishing on 2023-08-31. The panel continues for a further 8,453 bars, 3.44% of the sample. Across all four of its channels the print rate through October 2023 is exactly 0.000. The unlimited fill carried a value from the final August print across the following eight months, and—because the availability indicator was computed *after* the fill rather than before it—labelled every one of those bars as observed. Measured on the family: raw coverage 0.2197, indicator mean 0.6983.

Three further feeds terminate the same way. **vix**, **vix3m** and **vvix** each stop on 2024-02-12, leaving 2,745 to 2,747 dead bars apiece inside the evaluation region. We did not identify these until we constructed the calendar check described below, and no result in the earlier literature on this panel accounts for them.

4.2 Why One Ordering Error Defeated Three Guards

Computing the indicator after the fill is a single line’s worth of ordering, and it disabled every protection the pipeline already carried.

The impute-and-indicate median substitution never fired, because the forward fill had left no missing value to substitute for. The model could not distinguish a fresh print from an eight-month-old one, the indicator reading ≈ 1 in both cases. And the scale guard—which floors each feature’s rolling interquartile range at an expanding IQR taken over its

real values—computed that floor over the forward-filled constants, so the floor was itself degenerate.

That last failure is the one that converts a stale feature into a destructive one. Features are expressed in rolling-IQR units. A frozen series has no dispersion, so its rolling IQR collapses toward zero, and the division that standardises the feature is then a division by nearly nothing. The volatility-demand columns reach 2,260 rolling IQRs. The series had not moved; the denominator had vanished.

4.3 What It Cost

On the October 2023 window the ridge fit on the full 504-feature design attains QLIKE 6.37170. On the same window with the defect repaired it attains 0.12391, against a backbone of 0.12823 that is unaffected either way. The forecast was one to two orders of magnitude below the realised value, and 27 bars out of 27,376 carry 47% of the only catastrophic exogenous failure in the twenty-year sample.

Pooled over the whole panel the exogenous block’s era-8 deficit against the backbone moves from +0.03321 to −0.00332. Every genuine modelling effect reported in this paper lies between 0.0005 and 0.004.

The consequence for our own earlier conclusions is severe enough to state plainly. Model comparisons run on the uncorrected panel were scored partly on how far each design happened to *dilute* the corrupted channel. A design that contracts each channel’s twelve rungs to a single amplitude attenuates it by a factor of twelve, and duly appeared to win; the raw parameterisation carried it faithfully, and duly appeared to lose. On the corrected panel the ordering reverses. We had drafted a version of this paper around the former result. Section 13 records it and thirteen other withdrawn claims.

4.4 The Check That Catches It

The defect is recoverable from one comparison, which we now run as a gate on every built cache:

For each channel, the mean of the availability indicator the pipeline constructed must equal the fraction of rows on which the raw feed actually printed.

On the uncorrected panel this fails for 31 of 41 channels; on the volatility-demand family it reads 0.698 against 0.223. The check requires no model, no held-out data and no threshold tuning—the two quantities are estimates of the same number, and any gap is the bug or one like it. We recommend it as standard for any mixed-frequency panel in which exogenous series are filled.

We report four further artefact-level invariants alongside it—scale collapse measured over live rows, staleness as the longest run of exact repetition, imputation share, and finiteness with a magnitude summary. Table 2 gives the violation counts across build stages.

Table 2: Artefact invariant violations by build stage, 41 channels. Class A is the availability check above; class B is scale collapse over live rows; class D records channels more than half imputed, which is a property of the data rather than a defect and is reported rather than repaired.

	uncorrected	availability fix	composed fix
A availability honesty	31	0	0
B scale collapse	0	10	1
D imputation share	7	7	7
worst channel, rolling IQRs	2260.4	336.5	96.2

4.5 The Fix Is an Interaction

The repair has two halves, and neither works alone. This is the part we would have missed had we not been forced to measure each separately.

Recording availability before the fill—the honest indicator—changes no feature *value*. It corrects the bookkeeping and leaves the corrupted column exactly as it was. Estimating each feature’s location and scale from observed rows only, without also repairing the values, leaves the volatility-demand family unchanged at 2,260 rolling IQRs. Substituting a neutral median on unobserved rows *without* masked estimation is actively harmful: it replaces varying stale values with a constant, which collapses the rolling IQR of a densely-observed channel and takes `effspread_vwstock` from 17.6 to 932.6.

Composed, they take the worst channel from 2,260.4 to 6.7. Three separate attempts at this repair failed before we tested the two-by-two, each because it shipped one half.

A third component proved necessary once the first two were in place. With availability recorded honestly, a channel’s scale is estimated from observed rows only—and in a feed’s opening weeks there are few of those. The estimator accepted a median and IQR from as few as eight observations, so a quiet opening fortnight set the scale for everything that followed. Each newly-affected channel’s worst value sits 13 to 116 days after its feed’s *first* print: a feed-initiation transient, the same defect family as the termination bug at the opposite end of a feed’s life. Requiring 512 observations before a scale is trusted, and flooring the IQR at 1% of the largest scale the channel has shown, takes the worst channel from 336.5 to 96.2 and class-B violations from 10 to 1.

4.6 What This Implies for Practice

The general statement is that a fill policy is a modelling choice, and in this setting it is the dominant one. At any given bar, 36% of the exogenous design is not a measurement but an echo of an earlier one; on the sparsest channels the figure is 78%. A model handed those columns without an honest indicator is fitting the fill policy, not the market, and it cannot tell the difference because nothing in the design distinguishes them.

We therefore report the ordering of effects we actually measured, which inverts the usual emphasis: the representation of missing and stale data dominates, protocol choices come next, and the model class—the subject of most of the literature this paper engages with,

and of most of our own effort—comes last.

5 Methodology

This section details the full modeling pipeline: target construction, a three-stage causal preprocessing pipeline, two alternative feature engineering strategies, model specifications, a walk-forward backtesting protocol, and evaluation procedures.¹

5.1 Realized Volatility

The forecasting target is realized variance (RV) computed at the 30-minute bar level from 5-minute intraday log-returns. For the j -th 30-minute bar on trading day d , realized volatility is defined as

$$RV_{d,j} = \sum_{i=1}^M r_{d,j,i}^2, \quad (1)$$

where $r_{d,j,i}$ denotes the i -th 5-minute log-return within the bar and $M = 6$ is the number of non-overlapping 5-minute intervals per 30-minute bar. Because data are recorded over 24-hour trading days (including pre-market, regular, and after-hours sessions), each day contains 48 bars of 30 minutes each. For notational convenience, we index observations sequentially as RV_t where t enumerates consecutive 30-minute bars across all trading days. The forecasting objective is to predict the one-step-ahead realized volatility RV_{t+1} .

5.2 Data Preprocessing

Raw realized volatility and exogenous variables undergo a three-stage transformation pipeline applied in the following fixed order:

$$\text{Raw series} \xrightarrow{\text{Stage 1}} \text{Diurnal adjustment} \xrightarrow{\text{Stage 2}} \text{Semantic transform} \xrightarrow{\text{Stage 3}} \text{Rolling winsorization}. \quad (2)$$

The pipeline is designed to (i) remove predictable intraday patterns, (ii) stabilize distributional properties, and (iii) limit the influence of extreme observations. Importantly, at evaluation time only Stages 1 and 2 are inverted to recover raw-scale forecasts; Stage 3 (winsorization) is a lossy operation that is not reversed.

5.2.1 Stage 1: Diurnal Adjustment

Intraday volatility exhibits a well-documented U-shaped pattern, with elevated levels near the market open and close. To remove this deterministic seasonality, we divide each variable by a rolling time-of-day baseline. Let $s(t) \in \{1, \dots, 48\}$ denote the time-of-day slot for observation t . For non-negative variables such as RV_t , the diurnal baseline is

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \quad (3)$$

¹All transformations are applied in a strictly causal (non-anticipative) manner: every quantity computed at time t uses only information available at or before $t - 1$.

where $\mathcal{W}_t = \{j : j < t, s(j) = s(t)\}$ denotes the set of up to $W = 20$ most recent observations sharing the same time slot, with a minimum of 5 observations required before the baseline becomes valid. The one-period shift ($j < t$, not $j \leq t$) ensures strict causality. The adjusted series is then

$$\widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}. \quad (4)$$

For signed variables (e.g., autocovariance), the baseline is instead the rolling standard deviation within each slot, and the adjustment divides by this rolling standard deviation. Observations in the initial warm-up period (before 5 same-slot observations have accumulated) have their baseline set to 1, effectively passing through unadjusted; these observations are subsequently consumed by the training buffer and do not enter the test set.

5.2.2 Stage 2: Semantic Transforms

Following diurnal adjustment, we apply variance-stabilizing transforms chosen according to the distributional properties of each variable class:

- Squared-return and volatility measures (RV , bipower variation, turnover, effective spread): $x \mapsto \sqrt{x}$.
- Autocovariance: $x \mapsto \text{sign}(x)\sqrt{|x|}$.
- Cubic returns: $x \mapsto x^{1/3}$.
- Quartic returns: $x \mapsto x^{1/4}$.
- Implied volatility measures (VIX, VVIX, VIX3M): $x \mapsto \log(x)$.

These transforms compress the right tail and reduce heteroskedasticity, which is particularly beneficial for linear models. The square-root transform applied to RV is especially important: since models are trained on the $\sqrt{RV}$ scale, a bias correction (Duan’s smearing estimator, Section 5.6.1) is required when converting forecasts back to the raw variance scale.

5.2.3 Stage 3: Rolling Winsorization

After transformation, each series is clipped to rolling 5th and 95th percentile bounds computed over a trailing window of $W_{\text{clip}} = 240$ observations (approximately 5 trading days):

$$x_t^w = \text{clip}\left(x_t; q_{0.05}^{(t)}, q_{0.95}^{(t)}\right), \quad (5)$$

where $q_{\alpha}^{(t)}$ denotes the α -quantile of the 240-observation trailing window ending at $t - 1$. The one-period shift ensures that quantile bounds are computed from past data only. Because winsorization is a lossy compression, it cannot be inverted at evaluation time; forecasts are mapped back to the raw scale using only the inverse of Stages 1–2 (Section 5.6.1).

5.3 Feature Engineering

We consider two alternative strategies for constructing lagged features from the preprocessed series: a hand-crafted HAR lag structure and a data-driven PCA compression. Both strategies transform each variable’s history into a fixed-dimensional feature vector suitable for regression.

5.3.1 HAR Features

The Heterogeneous Autoregressive (HAR) model of Corsi [2009] captures the multi-scale persistence of volatility through aggregated lagged components. We adopt a geometric base-5 lag structure adapted to the 30-minute sampling frequency, yielding the lag set

$$\mathcal{L} = \{1, 5, 25, 125, 625, 3125\}, \quad (6)$$

which corresponds approximately to horizons of 30 minutes, 2.5 hours, half a trading day, 2.6 trading days, 13 trading days, and 65 trading days. The base-5 geometric spacing provides convenient multi-scale coverage at 30-minute resolution, analogous to the daily/weekly/monthly structure in Corsi [2009] but extended to higher frequencies. For each lag $k \in \mathcal{L}$, the HAR feature is the trailing rolling mean of the preprocessed target:

$$\overline{RV}_t^{(k)} = \frac{1}{k} \sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \quad (7)$$

where the shift by one ensures no contemporaneous information enters the feature. The standard HAR regression model takes the form

$$\widetilde{RV}_{t+1} = \beta_0 + \sum_{k \in \mathcal{L}} \beta_k \overline{RV}_t^{(k)} + \varepsilon_{t+1}. \quad (8)$$

When exogenous variables are included, the same HAR lag structure is applied to each additional preprocessed column, generating $|\mathcal{L}| = 6$ rolling-mean features per variable.

5.3.2 PCA Features

As an alternative to the fixed HAR lag structure, we consider a data-driven approach that serves two purposes: (i) it lets the data determine the optimal lag weighting rather than imposing the HAR rolling-mean structure, and (ii) it reduces dimensionality when many exogenous variables are included. We first construct approximately 20 log-spaced raw lags spanning the range $[1, 3125]$, providing dense coverage at short horizons and sparser coverage at longer horizons. For each variable, individual shifted lags $x_{t-\ell}$ for $\ell \in \mathcal{L}_{\text{pca}}$ are computed (rather than rolling means as in the HAR approach).

These raw lag features are then projected onto a low-dimensional subspace via Principal Component Analysis (PCA). Specifically, on each rolling training window of length T_{train} , we fit PCA with $n_c = 5$ components using a randomized SVD solver:

$$\mathbf{z}_t = \mathbf{W}^\top (\mathbf{x}_t - \bar{\mathbf{x}}), \quad (9)$$

where $\mathbf{W} \in \mathbb{R}^{p \times n_c}$ is the projection matrix estimated from the training buffer and $\bar{\mathbf{x}}$ is the training-set mean. The PCA transform is re-estimated each time the model is refit, ensuring the projection adapts to evolving autocorrelation structures.

5.3.3 Exogenous Variable Subgroups

To assess the marginal contribution of different information sources, we organize exogenous variables into thematic subgroups. Each subgroup augments the target-only baseline (HAR or PCA features of $\widetilde{RV}$ alone) with additional features derived from the subgroup’s variables using the same lag structure. The subgroups are:

- **Baseline:** HAR (or PCA) features of $\widetilde{RV}$ only — no exogenous variables.
- **Moments:** Higher-order return moments and related measures (bipower variation, realized semi-variance, autocovariance, absolute returns, cubic and quartic returns).
- **Liquidity:** Order flow and market microstructure variables (volume, turnover, buy/sell turnover, effective spread, quoted spread, number of observations).
- **Market (EW):** Equal-weighted cross-sectional stock factor aggregates (returns, bipower, semi-variance, absolute returns).
- **Market (VW):** Value-weighted analogues of the equal-weighted factors.
- **Sentiment:** Social media sentiment metrics (StockTwits attention, sentiment score, sentiment count).
- **Implied Volatility:** Options-implied measures (VIX, VVIX, VIX3M).
- **Volatility Demand:** Options order flow measures (SPX open/close, all options open/close, open only).
- **All Features:** The union of all exogenous variables above.

5.4 Models

We consider five models spanning parametric linear, ensemble tree-based, and naive approaches. All models are trained on the preprocessed (diurnally adjusted, transformed, winsorized) scale. Every arm is refit at every bar. For the linear arms this is made tractable by the exact rank-1 and homotopy updates of Section 9.2; for the tree arms it is a deliberate expense, incurred so that refit cadence cannot confound the model-class comparison of Section 8.

5.4.1 Naive Baseline

The naive baseline predicts $\widetilde{RV}_{t+1}$ using the HAR rolling mean at the 125-period horizon (approximately 2.6 trading days):

$$\hat{y}_{t+1}^{\text{naive}} = \overline{RV}_t^{(125)}. \quad (10)$$

This provides a non-trivial benchmark that reflects the strong persistence of realized volatility.

5.4.2 Ridge Regression

Ridge regression augments ordinary least squares with an ℓ_2 penalty on the coefficient vector:

$$\hat{\beta} = \arg \min_{\beta} \{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \|\beta\|_2^2 \}, \quad (11)$$

with regularization parameter α re-selected causally (Section 5.4.6); the shared incumbent is the $\alpha = 0$ limit of this path, i.e. exact per-bar OLS. Input features are standardized using a rolling robust scaler based on the median and interquartile range (IQR) estimated from the current training buffer. Ridge is refit at every time step via the closed-form solution.

5.4.3 XGBoost

We employ XGBoost [Chen and Guestrin, 2016] as a gradient-boosted tree ensemble with the histogram-based splitting method. Hyperparameters are re-selected causally by the protocol of Section 5.4.6; the learning rate is pinned at the production value of 0.1 and the search ranges over max-depth and column subsampling. The model is refit at every bar.

5.4.4 LightGBM

LightGBM [Ke et al., 2017] is estimated under the same causal selection protocol and search grid as XGBoost, and is likewise refit at every bar.

5.4.5 Random Forest

A random forest ensemble is estimated under the same causal selection protocol. It does not appear in the frozen battery of Sections 7–9 and is retained here only for completeness of the model description.

5.4.6 Hyperparameter Selection

All hyperparameters reported in this paper are selected by the *causal* rolling protocol of Section 9.1: every 250 solves, each arm re-selects its configuration on a leakage-clean forward split of its own current training window (fit block, 25-bar embargo covering the longest moving average, 125-bar validation tail scored by mean squared error). No forecast’s hyperparameters are selected using information from that forecast’s future, and the selection loss is never the evaluation loss.

A superseded protocol, retained as a cautionary result. An earlier version of this study instead performed Bayesian optimization with Optuna [Akiba et al., 2019] whose objective was *QLIKE on the full walk-forward backtest* — that is, on the evaluation panel itself, under the evaluation loss. Trial counts also differed sharply across models, so arms were ranked partly by how many draws each had taken from the evaluation set. We report the magnitude of the resulting distortion in Section 8.4 because it calibrates how large a protocol artifact can be on this task; no number produced under that protocol appears elsewhere in this paper.

Categorical time features. All three tree-based models (XGBoost, LightGBM, random forest) additionally receive day-of-week and hour-of-day as categorical feature columns appended to the HAR or PCA features. Ridge regression does not receive these features, as the diurnal adjustment (Stage 1) already removes time-of-day effects from the input series.

5.5 Walk-Forward Backtesting

All models are evaluated using a strictly out-of-sample walk-forward procedure. At each test period t , the model is trained (or updated) on a rolling window of T_{train} consecutive observations immediately preceding t . **Every result reported in this paper uses $T_{\text{train}} = 24,000$ bars** (approximately 500 trading days at 48 bars per 24-hour day), with model refit at every bar for every arm, so that no refit-cadence asymmetry confounds the model-class comparison. A shorter $T_{\text{train}} = 500$ window and a five-step tree refit cadence were used in superseded experiments; those settings produce none of the numbers reported here and are noted only because earlier internal tables used them. The procedure is:

1. **Initialize:** Fit the model on observations $\{t - T_{\text{train}}, \dots, t - 1\}$.
2. **Predict:** Generate forecast $\hat{y}_t$ using features $\mathbf{x}_t$.
3. **Observe:** Record the realized target y_t .
4. **Update:** Add $(\mathbf{x}_t, y_t)$ to the rolling buffer, drop the oldest observation, and refit the model at the model-specific refit frequency (every step for ridge, every 5 steps for tree models).
5. **Advance:** Increment t and return to step 2.

For ridge regression, feature scaling parameters (median and IQR) are updated at every step using the rolling robust scaler fit on the current training buffer. For PCA-based configurations, the projection matrix $\mathbf{W}$ is re-estimated at each refit. This ensures all model components adapt to non-stationarities while maintaining strict temporal separation between training and test data.

The backtest is parallelized across 100 chunks via SLURM array jobs, with each chunk processing a contiguous subset of test indices.

5.6 Evaluation

5.6.1 Inverse Transform and Duan’s Smearing Estimator

Models are trained on the preprocessed scale (after diurnal adjustment, semantic transform, and winsorization). To evaluate forecasts against raw realized volatility, predictions must be mapped back through the inverse of the preprocessing pipeline. Of the three stages, only Stages 1 (diurnal adjustment) and 2 (semantic transform) are invertible; Stage 3 (winsorization) is lossy and is not reversed.

For the primary target RV , the semantic transform is the square root ($\sqrt{\cdot}$), so the naïve inverse would simply square the forecast: $\hat{y}_t^2$. However, because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$ by Jensen’s

inequality, squaring a forecast of $\sqrt{RV}$ produces a downward-biased estimate of RV . Duan’s smearing estimator [Duan, 1983] corrects this bias by adding the in-sample residual variance before inverting:

$$\widehat{RV}_t^{\text{raw}} = (\hat{y}_t^2 + \hat{\sigma}_\varepsilon^2) \cdot \bar{RV}_{t|s}^{\text{baseline}}, \quad (12)$$

where $\hat{\sigma}_\varepsilon^2 = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2$ is the in-sample mean squared residual and $\bar{RV}_{t|s}^{\text{baseline}}$ is the diurnal baseline factor at time t that reverses Stage 1. Equation (12) is the analytic form of this correction rather than Duan’s empirical smearing average $\frac{1}{N} \sum_i (\hat{y}_t + \hat{\varepsilon}_i)^2$; the two coincide when the residuals are mean-zero, since $\mathbb{E}[(\hat{y} + \varepsilon)^2] = \hat{y}^2 + 2\hat{y}\mathbb{E}[\varepsilon] + \sigma_\varepsilon^2$. Two biases are left uncorrected and we do not quantify them: $\hat{\sigma}_\varepsilon^2$ is estimated on the winsorized scale while the evaluation target is unwinsorized, and Stage 3 is not inverted at all (Section 14).

5.6.2 QLIKE Loss

Our primary evaluation metric is the quasi-likelihood (QLIKE) loss function, which is the robust loss function for volatility forecast comparison under general loss functions [Patton, 2011]:

$$L_{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^N \left[\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} - \log \left(\frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}} \right) - 1 \right]. \quad (13)$$

QLIKE is a strictly consistent scoring rule for the conditional variance, and its practical appeal in this literature is robustness: Patton [2011] shows that the ranking it induces is preserved when the target is a noisy but conditionally unbiased proxy for integrated variance, which a realized measure is. It penalizes under-prediction more heavily than over-prediction. That asymmetry is not neutral for model comparison — Section 8.6 shows that on this panel QLIKE systematically prefers attenuated forecasts — and Section 8.5 reports that QLIKE and raw-variance MSE rank our battery almost independently.

5.6.3 Additional Metrics

We also report mean squared error (MSE) and mean absolute error (MAE) computed on the preprocessed (adjusted) scale. To limit the influence of extreme outliers, winsorized variants (clipping errors to the 5th–95th percentile range) are computed for all metrics. Out-of-sample R^2 is defined relative to the naive baseline:

$$R_{\text{OOS}}^2 = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{naive}}}, \quad (14)$$

where positive values indicate improvement over the naive forecast.

5.7 Experimental Design

The paper reports one frozen battery of 98 arms on one panel of $n = 218,934$ bars, with identical rows for every cell, organised into four analyses that each vary one thing at a time:

1. **Lag-ladder sweep** (Section 6.3). The estimator (OLS), the information set (backbone only) and the panel are held fixed while the geometric base and truncation cap of the lag ladder vary. This isolates quadrature error in the backbone itself.

2. **Bucket attribution** (Section 7). The model class is held fixed at causally tuned ridge while the exogenous information set varies over the eight buckets and their union, so that information is never confounded with capacity.
3. **Model-class comparison** (Section 8). The information set is held fixed while the estimator varies over five classes, all causally tuned, all refit every bar.
4. **Algorithm ablations** (Section 9). The feature set is frozen entirely and the estimation algorithm varies, including the matched embedding ablations of Table 10.

Superseded analyses — the variable- N subgroup tables and the evaluation-panel hyperparameter search — contribute no number to this paper; Section 14 explains why they were withdrawn.

6 Descriptive Analysis: The Case for a Kernel Backbone

Before asking what exogenous information adds, we fix the instrument that will measure it. This section argues that the target’s own autoregressive structure determines that instrument, and determines it as a *truncated power-law convolution kernel modulated by a session clock*—of which the standard HAR regression is a coarse quadrature. The argument matters for the rest of the paper because every attribution in Sections 7 and 8 is measured as an increment over this backbone, and an incorrectly specified backbone donates its own approximation error to whatever is being credited.

6.1 The Panel and the Dense-but-Weak Fact

The design matrix has 529 columns. Its covariance concentrates in a handful of directions, but the forecasting signal does not live in those directions. Three measurements establish this. They predate the frozen battery and were made on the earlier all-buckets cache; we report them because the regime they describe is what the rest of the paper is about, and we mark the levels that are not panel-comparable:

- **Low-rank compression destroys skill.** Principal-component truncation of the exogenous block retains most of the variance ($K = 5$ recovers 72%, $K = 20$ recovers 90%) but loses to dense shrinkage at every penalty setting; the best PCA-20 configuration reaches QLIKE 0.12890 against 0.12530 for the dense elastic net on the same cache. The signal is not in the leading eigendirections. *Panel note:* these two levels come from the earlier all-buckets cache, not the frozen battery, and are comparable to each other but not to Table 9. **[pending: PCA/PLS truncation arms on the frozen panel]**
- **ℓ_1 selection recovers a small unstable core plus a wide shallow tail.** Elastic net at its optimum retains roughly 98–120 of 529 columns; pushing sparsity harder costs performance monotonically.

- **No individual column is strong.** The per-feature effects are uniformly small, and the strongest single engineered feature contributes less than the aggregate of the tail it sits in.

We use *dense but weak* throughout as shorthand for this configuration: sparse in no useful coordinate basis, not low-rank in data space, and with no individually decisive column. Section 8 shows what this implies for model class; the immediate consequence is that the backbone cannot be discovered by selection and must be specified.

6.2 HAR Is a Quadrature Rule

Write the general one-step linear forecast as a convolution of the target’s own past against a decay kernel φ :

$$\hat{y}_t = \mu + \sum_{u \geq 1} \varphi(u) y_{t-u}. \quad (15)$$

HAR does not estimate φ freely. It regresses on trailing rolling means (boxcars) over a ladder of rungs r_k :

$$\hat{y}_t = \mu + \sum_k w_k \cdot \frac{1}{r_k} \sum_{u=1}^{r_k} y_{t-u}. \quad (16)$$

Collecting the total coefficient that Equation (16) places on an individual lag u gives the *effective kernel* implied by a fitted HAR:

$$\boxed{\varphi(u) = \sum_{k: r_k \geq u} \frac{w_k}{r_k}} \quad (17)$$

which is a decreasing step function with breakpoints at the rungs. Under this reading the $|\mathcal{L}|$ fitted HAR coefficients are not $|\mathcal{L}|$ distinct facts about volatility at $|\mathcal{L}|$ distinct horizons: they are quadrature weights on a piecewise-constant approximation to a single underlying curve, with the rungs as knots. Figure 1 plots the fitted boxcar coefficients and the implied φ .

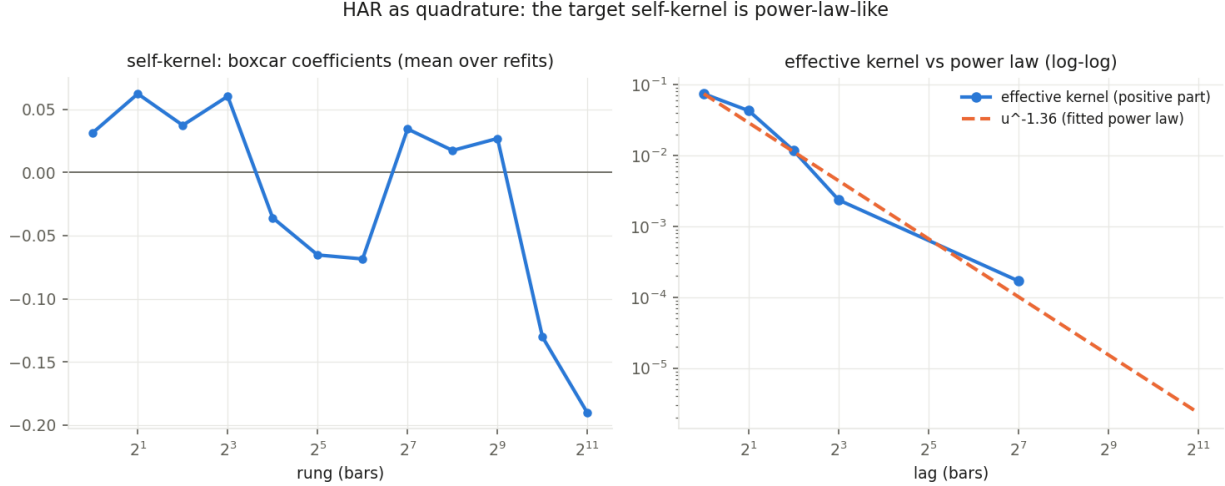


Figure 1: Left: self-kernel boxcar coefficients, averaged over walk-forward refits. Right: the implied effective kernel of Equation (17) on log-log axes, against a fitted power law. The straight-line behaviour on log-log axes is the object this section is about. Only the positive part of φ is plotted; Section 6.5 discusses the negative lobes, which are not an incidental detail.

6.3 The Kernel Is Power-Law and Truncated

The implied kernel is approximately linear on log-log axes, and a single-parameter power law $\varphi(u) \propto u^{-\hat{\beta}}$ approximately reproduces the entire fitted ladder—the strongest available evidence that the ladder approximates one curve rather than measuring several components.

Earlier drafts quoted $\hat{\beta} \approx 1.36$. That figure was a plotting constant, never an estimate. Estimating it changes the number and, with it, one of the paper’s conclusions. The self-kernel is a poor place to estimate from: only four of its twelve lags are positive (Section 6.5), and fitting the log-log slope on those four gives $\hat{\beta} = 1.51$ with a standard error of 0.54—a 95% interval of $[0.45, 2.56]$ that contains every value any competing theory proposes.

The exogenous cross-kernels are a far better instrument, for a specific reason: a log-log slope is invariant to the arbitrary scaling of each channel, so the 41 channels estimated in each of six session bins give 244 comparable estimates of the same shape parameter, with a median of six positive lags each. Clustering by channel—the six session fits for one channel come from one series and are not independent—gives

$$\hat{\beta} = 1.033, \quad \text{SE} = 0.036, \quad 95\% \text{ CI } [0.963, 1.104]. \quad (18)$$

This rejects the 1.36 of earlier drafts ($t = -9.1$, $p = 2.6 \times 10^{-11}$), is consistent with the ≈ 1 reported throughout the Hawkes-in-finance literature ($t = +0.9$, $p = 0.36$), and rejects the ≈ 1.6 that the rough-volatility correspondence would imply ($t = -15.8$, $p = 8.4 \times 10^{-19}$).

We report one disagreement rather than suppress it. The *marginal* self-kernel, correctly conditioned and on the raw basis (Section 6.6), gives $\hat{\beta} = 1.506$ with a standard error of 0.035 over ten positive lags—materially steeper than the cross-kernel estimate above, and rejecting both 1.0 and (at 2.7 standard errors) the 1.6 a rough-volatility correspondence implies. The two estimates are different objects: own-lag response versus cross-channel response. Both

are well identified, so the gap is a fact about the data rather than an artefact, and it means the paper reports two exponents rather than one.

If the kernel is genuinely power-law—that is, scale-free—then the quadrature reading of Section 6.2 makes three predictions that are not made by the conventional reading of HAR as a daily/weekly/monthly cascade. All three are testable on the shared panel, and all three are confirmed.

Prediction 1: a scale-free kernel wants a scale-free grid, and refining the grid pays. Uniform knots in log-lag are the natural quadrature for a power law, and halving the geometric base doubles the knot density per decade. Error should therefore fall monotonically as the base falls. Table 3 reports the measured ladder.

Table 3: HAR ladder base sweep: OLS on a geometric lag ladder with base b , all arms on the shared panel ($n = 218,934$), all differences measured against the same incumbent. “explicit” is the conventional daily/weekly/monthly HAR lag structure of Corsi [2009] transposed to the 30-minute grid. Within each base we report the best cap; the cap is a near-non-lever (within-base spread ≈ 0.0001) while the base is not (across-base spread ≈ 0.0014). Negative DM t favours the ladder arm.

Ladder base	Arm	QLIKE	DM t vs. incumbent
$b = 2$	b2_c3125	0.13332	−10.58
$b = 3$	b3_c3125	0.13349	−10.19
$b = 4$	b4_c3125	0.13361	−8.47
$b = 5$	b5_c3125 (incumbent)	0.13415	—
$b = 6$	b6_c3125	0.13433	+3.12
explicit	explicit_c3125	0.13439	+3.10
$b = 8$	b8_c3125	0.13476	+7.31

The ordering is monotone in the base across all six bases—the rank correlation between base and QLIKE is exactly +1.000, with no inversions—and DM $t = -10.6$ for $b=2$ against the incumbent $b=5$. The signature is precisely that of a quadrature error: *how finely the kernel is sampled matters, where the sum is truncated barely does*. This is difficult to explain if the rungs were recovering genuinely distinct daily, weekly and monthly components, and easy to explain if they are knots on one smooth curve. We note that the conventional explicit HAR structure is not merely suboptimal but *worse than the incumbent*, at DM $t = +3.1$.

Prediction 2: power law, not exponential. A single exponential kernel fitted in place of the power law is +0.011 QLIKE worse—roughly an order of magnitude larger than any effect elsewhere in this paper. The kernel shape agrees: fitting $\log \varphi$ against $\log u$ (power law) and against u (exponential) on the same points gives $R^2 = 0.80$ and 0.44 respectively, with the power law preferred by AIC. This is the one prediction of the three that the fitted shape supports without qualification.

Prediction 3: truncation is real, and it is an interior optimum. Direct power-law convolution features degrade sharply when the convolution window is extended too far, while the boxcar ladder’s cap is a near-non-lever. Per-bar Diebold–Mariano tests on the campaign tile (Section 11.3) resolve the direct-feature side of this in both directions: a truncation at $U = 256$ beats $U = 32$ by 0.0074 ($t = -4.16$), beats $U = 128$ by 0.0026 ($t = -3.50$), and beats no truncation at all by 0.0077 ($t = -3.83$), every comparison surviving Holm adjustment across the contrast family. Truncating the kernel is therefore not a computational convenience—removing it costs real accuracy—and the optimum is interior rather than at either extreme.

We attribute the contrast with the boxcar ladder’s indifference to its cap to normalisation: the direct features normalise weights to sum to one, so extending U redistributes mass away from recent lags, whereas adding a boxcar rung leaves the other rungs undisturbed. That reconciliation remains an argument rather than a measurement.

6.4 The Hawkes Reading

Equation (15) with a power-law φ is, in discrete time, the conditional-intensity equation of a self-exciting point process [Hawkes, 1971]. This subsection makes the correspondence explicit, states what it buys, and—at equal length—states where it does not yet hold.

6.4.1 From point process to variance regression

Let N be a linear Hawkes process with conditional intensity

$$\lambda(t) = \mu + \int_{-\infty}^t \varphi(t-s) dN(s), \quad \varphi \geq 0, \quad (19)$$

and let the price be a pure-jump process driven by N . Realized variance over a bar is then, up to the squared jump size, the event count over that bar, so that $RV_t \propto N(t - \Delta, t)$ and

$$\mathbb{E}[RV_t | \mathcal{F}_{t-1}] \propto \int_{\text{bar}} \mathbb{E}[\lambda(s)] ds = \tilde{\mu} + \sum_{u \geq 1} \bar{\varphi}(u) RV_{t-u}, \quad (20)$$

where $\bar{\varphi}$ is φ aggregated to the bar grid. Equation (20) is Equation (15). The correspondence is therefore not an analogy: for a linear Hawkes process the conditional intensity is exactly linear in the observed past, so a regression of counts on past counts identifies φ itself rather than a proxy for it. What we estimate is a bar-aggregated Hawkes kernel, subject to the caveats of Section 6.5.

6.4.2 The branching ratio is already estimated

The central quantity of Hawkes theory is the branching ratio $n = \int_0^\infty \varphi(u) du$: the expected number of direct offspring per event, equivalently the fraction of activity that is endogenously generated. Stationarity requires $n < 1$, and $n \rightarrow 1$ is the “critical reflexivity” regime that has been the subject of an extended empirical debate [Hardiman et al., 2013, Filimonov and Sornette, 2012].

For the boxcar parameterisation of Equation (16) there is an exact identity. Substituting Equation (17),

$$\sum_{u=1}^{r_{\max}} \varphi(u) = \sum_u \sum_{k: r_k \geq u} \frac{w_k}{r_k} = \sum_k \frac{w_k}{r_k} |\{u : 1 \leq u \leq r_k\}| = \sum_k w_k. \quad (21)$$

The sum of the fitted HAR coefficients is the discretised kernel integral. We verify the identity numerically to machine precision (-0.038000 against -0.038000 , error 2.5×10^{-15}), so every fitted ladder yields a branching-ratio analogue at no additional computational cost.

Computing it is not kind to the structural reading. **[pending: restore the pooled/by-session $\hat{n}$ table (tab:hawkes_kernel)]** The pooled value is $\hat{n} = -0.038$: **negative**, and therefore outside the admissible range $[0, 1)$ entirely rather than merely sub-critical. Across sessions it ranges from $+0.020$ overnight to -0.260 in the after-hours session, so its *sign* is not even stable. Whatever this quantity is measuring, it is not a branching ratio, and we do not report a reflexivity estimate. Section 6.5 argues that the estimation design, not the market, is the reason.

We flag one consistency point in our favour. The reflexivity debate turns on whether high estimates of n are artifacts of an unmodelled time-varying background intensity $\mu(t)$: Filimonov and Sornette [2012] argue that ignoring intraday seasonality biases $\hat{n}$ upward toward one. Our Stage-1 diurnal adjustment (Section 5.2.1) is exactly the $\mu(t)$ correction that this critique demands, adopted originally for an unrelated reason. Section 6.5 argues that it may be too aggressive in the other direction.

6.4.3 Two existing negative results become predictions

The Hawkes reading retro-predicts two of the paper’s otherwise awkward findings.

Bounded nonlinearity. The quadratic Hawkes model of Blanc et al. [2017] makes the feedback a quadratic form in the past rather than a linear functional, generating volatility clustering and fat tails endogenously and capturing the Zumbach effect that linear ARCH-type feedback misses. What it predicts about extractable structure is specific: real nonlinearity, *additive plus pairwise, and no more*. That is precisely the saturation reported in Section 8.7—the recoverable nonlinearity is additive-plus-pairwise anchored to the session clock, and deeper capacity does not pay. A result that reads as a puzzling ceiling under a generic machine-learning framing reads as a confirmation under this one.

A low-rank operator. A multivariate Hawkes process has a matrix kernel $\Phi_{ij}(u)$, and a low-rank factorisation $\Phi_{ij}(u) \approx \sum_k a_k u_i^{(k)} v_j^{(k)} g_k(u)$ is the natural structural restriction on it. The operator result—41 exogenous channels sharing a small number of lag profiles, with quasi-static geometry and fast amplitude motion—is such a factorisation, and it is measurable directly. Across the six session bins, the 41×12 exogenous operator needs only **4 to 6** singular components to carry 90% of its energy, with the leading component taking 28–58%. The rank-5 structure the paper reports is therefore a property of the operator and not of any particular estimation window. Nonparametric estimation of multivariate Hawkes

kernels and their low-rank structure is an established problem [Bacry et al., 2015a, Achab et al., 2017], which gives this part of the analysis a comparison class it currently lacks.

We attach the same caveat as above, and it is a serious one for this reading: close to half of the cross-kernel cells are negative, so the operator is a contrast—each channel compared against a reference past—rather than a matrix of excitation kernels. The low-rank finding is robust; its interpretation as a multivariate Hawkes kernel matrix is not.

6.5 Where the Reading Does Not Yet Hold

Three discrepancies stand between the correspondence above and a defensible structural claim. We state them here rather than in a limitations section because two of them plausibly share a single cause, and that cause is our own preprocessing.

Table 4: Kernel diagnostics by session. $\hat{n} = \sum_k w_k$ is the discretised kernel integral of Equation (21), the branching-ratio analogue; a linear Hawkes process requires $\hat{n} \in [0, 1)$. “Neg. lags” counts lags at which the effective kernel is negative, which a linear Hawkes kernel cannot be, and “neg. mass” is their share of total absolute kernel mass. $\hat{\beta}$ is the log-log slope over the positive part, with the textbook OLS standard error; because the kernel values are nested partial sums of one coefficient vector, that error understates the true uncertainty. cos is the similarity of the session kernel to the pooled one.

Session	Bars	$\hat{n}$	Neg. lags	Neg. mass	$\hat{\beta}$	SE	cos
pooled	242,934	−0.0380	8/12	10.3%	1.51	0.54	—
overnight	80,650	+0.0200	6/12	6.6%	1.25	0.12	+0.83
early	54,758	−0.0200	5/12	11.8%	1.04	0.08	+0.05
open	14,934	+0.0000	7/12	87.8%	1.15	0.14	−0.55
midday	34,846	−0.0500	7/12	61.8%	1.18	0.24	+0.32
close	14,934	−0.1300	8/12	56.9%	0.47	0.21	−0.20
after	42,812	−0.2600	8/12	86.3%	1.11	0.60	+0.31

(i) The archived kernel takes negative values. A linear Hawkes kernel must be non-negative; Equation (19) does not otherwise define a point process. On the archived coefficient vector, φ violates this badly: 8 of 12 lags are negative, carrying 10.3% of total absolute kernel mass, rising to 87.8% at the open and 86.3% after hours. The archived exogenous cross-kernels are no better—between 44.7% and 50.4% of all (channel, lag) cells are negative, indistinguishable from a coin flip. The archived integrated kernel is $\hat{n} = -0.038$, outside the $[0, 1)$ range a linear Hawkes process admits rather than merely sub-critical, and its sign is not stable across sessions.

(ii) None of it belongs to the kernel. Section 6.6 rebuilds the feature cache, reproduces the archived vector exactly, and then isolates the cause: the archived estimate conditions the self-kernel on 41 exogenous channels, several of which are direct volatility measures. The *marginal* self-kernel on the same rows carries zero negative mass and an integrated kernel of

+0.913. Every diagnostic in this subsection is therefore a property of a partial effect between correlated volatility proxies, not of the target’s own kernel.

6.6 The Kernel, Rebuilt and Correctly Conditioned

The diagnostics above are computed on a coefficient vector recovered from an archived driver log. We rebuilt the feature cache that produced it—`run_geometry_local.prepare_full('all_features_har_base=2')`, the build documented in the campaign README—and reran the archived probe against it. Three things follow, and they overturn the reading in Section 6.5.

The estimate reproduces exactly. The rebuilt cache returns the archived coefficient vector to three decimal places (correlation +1.0000, maximum absolute difference 0.0005, $\hat{n}$ −0.0375 against −0.0380). The pipeline is reproducible; nothing about the archived number is an accident of a lost artefact.

The archived vector is not on the kernel’s basis. `prepare_full`’s final step is a rolling robust scaler: column j at row t becomes $(x - \text{median}_{[t-W,t)})/\text{IQR}_{[t-W,t)}$. A coefficient on that is a per-IQR coefficient, whereas the identity $\sum_u \varphi(u) = \sum_k w_k$ is stated for coefficients on the raw boxcar means. Converting back—dividing each coefficient by its column’s mean scaling IQR—is a prerequisite for reading the vector as a kernel at all. It is not, however, the explanation for the sign structure: on the raw basis the archived integrated kernel is −0.363, further from admissible rather than closer.

Table 5: Nested probes on the rebuilt `b2_mmap` cache, all on the same 242,934 rows. Coefficients are converted from the cached rolling-robust-scaled basis to the raw boxcar basis (dividing by each column’s mean scaling IQR) so that $\hat{n} = \sum_k w_k$ is the discretised kernel integral. Adding the exogenous ladder—which contains bipower variation, absolute returns and semivariance, all near-substitutes for lagged RV —is what flips the self-kernel’s sign.

Probe	Conditioning set	Cols	$\hat{n}$	Neg. lags	Neg. mass	$\hat{\beta}$
A	marginal self-kernel	12	+0.913	2/12	0.0%	1.51
B	+ 41 exogenous ladders	504	−0.139	6/12	16.4%	1.01
D	+ availability indicators	996	−0.153	6/12	17.6%	0.96
C	+ calendar and activity flags	585	−0.264	7/12	13.6%	1.34
E	the archived probe	1077	−0.363	9/12	14.4%	1.35

Conditioning on volatility proxies is the explanation. [pending: restore the nested-probe table (tab:nested_probes)] We fit nested probes on the rebuilt cache, all on identical rows, all converted to the raw basis. The marginal self-kernel—own lags only, the object a Hawkes branching ratio actually denotes—is *well behaved on every diagnostic*:

$$\hat{n} = 0.913, \quad \text{negative mass } 0.0\%, \quad \hat{\beta} = 1.506 \text{ (SE } 0.035, \text{ 10 positive lags)}, \quad (22)$$

with φ monotonically decreasing across a decade and a half of lags (0.405, 0.137, 0.052, 0.022, 0.007, ...). Adding the 41 exogenous ladders alone flips it to $\hat{n} = -0.139$ with 16.4% negative mass; the full archived conditioning set takes it to -0.363 .

The mechanism is not subtle once stated. The exogenous block contains bipower variation, absolute returns, realized semivariance and quarticity—these are *volatility measures*, near-substitutes for lagged RV rather than independent information. Conditioning on them absorbs the target’s own persistence, and what remains on the HAR coefficients is a contrast between correlated volatility proxies, not a self-excitation kernel. The negative lobes and the inadmissible integral are properties of that contrast.

What this rehabilitates, and what it does not. The marginal kernel supports the self-exciting reading rather than embarrassing it: an integrated kernel of 0.913 is sub-critical and close to unity, which is where the reflexivity literature places financial activity [Hardiman et al., 2013, Filimonov and Sornette, 2012], and it is achieved with no negative mass and a clean power-law decay. Read as an endogeneity share, it says roughly 91% of intraday variance dynamics is self-generated.

Two caveats keep this short of a structural claim. First, $\hat{n}$ is estimated on diurnally adjusted, square-root transformed, winsorized data with a ridge penalty, all of which bias it; the number is an analogue, not an identified branching ratio. Second, the objection of Section 6.4 stands regardless: we observe 30-minute aggregated variance, not events, so no quantity here is a point-process branching ratio in the literal sense. **[pending: estimate the marginal kernel without the ridge penalty and without winsorization, and bootstrap $\hat{n}$, before quoting an endogeneity share]**

Correction of record. Earlier drafts of this section reported the archived vector’s $\hat{n} = -0.038$ as an inadmissible branching ratio and drew structural conclusions from it. That number is a partial effect on a scaled basis; both the estimand and the basis were wrong. Neither the diurnal filter nor the winsorization nor the availability indicators is implicated—the 2×2 that tested the first two exonerated them, and the nested-probe sequence (**[pending: tab:nested_probes]**) isolates the exogenous block as the cause. We report the sequence because it is a general hazard: a self-kernel estimated jointly with volatility proxies is not a self-kernel.

6.7 The Properly Specified Object: A Multiplicative Error Model

Everything above estimates the kernel by unconstrained regression on a transformed target. That is the wrong tool for a non-negative observable, and it is why non-negativity had to be *tested* rather than guaranteed. The right object for aggregated non-negative data is the multiplicative error model [Engle and Gallo, 2006], equivalently INGARCH, which is the self-exciting intensity equation written for a continuous non-negative variable rather than a counting measure:

$$RV_t = \lambda_t \varepsilon_t, \quad \varepsilon_t \geq 0, \quad \mathbb{E}[\varepsilon_t] = 1, \quad \lambda_t = \omega + \sum_k \alpha_k \overline{RV}_{t-1}^{(r_k)}, \quad \alpha_k > 0. \quad (23)$$

Three properties make this the correct estimand. Non-negativity is imposed rather than tested, so the negative-lobe question cannot arise. The branching ratio is a parameter, $n = \sum_k \alpha_k$, with a standard error rather than a sum of unconstrained coefficients. And it is fitted on the raw variance scale, so no square-root transform, smearing correction or winsorisation intervenes. Convenient for this paper, the exponential quasi-likelihood whose log-density is $-(\log \lambda_t + RV_t/\lambda_t)$ makes QMLE estimation exactly QLIKE minimisation—the model is fitted under the paper’s own loss.

Seasonality is a *fixed* per-slot factor estimated on the training split, not the rolling divisor used elsewhere in this project. A fixed seasonal cannot absorb persistence, which removes the confound of Section 14.

Table 6: Branching ratio under three specifications, all on the same 242,462 bars with a 60/40 split. The MEM arms impose $\alpha_k > 0$ and are fitted by QMLE on the raw variance scale; standard errors are from the observed information with the delta method for n . The ladder regression is the unconstrained sqrt-scale ridge fit of Section 6.6, shown for comparison.

Specification	n	SE	OOS QLIKE	Constraint
HAR-MEM (base-2 ladder)	0.9917	0.0043	0.24184	$\alpha_k > 0$ imposed
MEM(1,1)	0.9753	0.0042	0.25707	$\alpha, \beta > 0$ imposed
Ladder regression (sqrt, ridge)	0.9133	—	—	unconstrained

The market sits at criticality. The constrained estimate is $n = 0.9917$ with a 95% interval of $[0.9834, 1.0001]$ —*statistically indistinguishable from unity*. The MEM(1,1) agrees at 0.9753 ± 0.0042 (implied persistence $\alpha + \beta = 0.9894$). Both sit above the unconstrained ladder’s 0.9133, which is what one expects when the constraint binds: forcing $\alpha_k > 0$ removes the negative lobes that were cancelling positive mass. These values land on the $n \approx 0.99$ that Hardiman et al. [2013] report for futures and read as critical reflexivity.

We decline to quote a cascade multiplier. At $n = 0.9917$ the point estimate $1/(1 - n)$ is 121, but at the interval’s lower end it is 60 and at its upper end it diverges. Near criticality the multiplier is not identified by data of this length, and reporting it would convey precision that does not exist.

The ladder beats the single decay. HAR-MEM reaches out-of-sample QLIKE 0.24184 against MEM(1,1)’s 0.25707. A single exponential recursion cannot reproduce the response that twelve non-negative ladder weights can, which is the same conclusion the exponential-versus-power-law comparison of Section 6.3 reached by a different route.

A model-free cross-check. Variance-ratio scaling— $\text{Var}(\sum_{i=1}^T RV_i) \propto T^{2H}$, estimated by regressing log variance on $\log T$ over $T = 1 \dots 1024$ bars—requires no model at all. It gives $H = 0.897$ with a standard error of 0.007, far above the $H = 0.5$ of a short-memory process. Strong long memory is what a near-critical kernel produces, so the model-free statistic and the constrained MEM agree.

We flag one conflation this invites. The H estimated here is the *long-memory* exponent of the level series’ partial sums. It is not the Hurst parameter of the rough-volatility literature,

which measures the Hölder regularity of log-volatility *paths* and is typically reported near 0.1. The two are different parameters and can coexist—rough at short scales, long-memory at long scales—which is precisely the decoupling Bennedsen et al. [2022] formalise. Our $H = 0.897$ therefore neither confirms nor contradicts $H \approx 0.1$; they are answers to different questions.

What this settles. The branching ratio is no longer an analogue. It is a parameter of a correctly specified, non-negativity-constrained model of a non-negative observable, estimated on the raw scale under the paper’s own loss, with a standard error. Its value says roughly 99% of intraday variance dynamics is endogenous. The residual caveats are honest but narrow: λ_t is a conditional mean for aggregated variance rather than a point-process intensity, so this is a self-excitation parameter for the variance process itself and not a literal offspring count; and near-critical estimates are known to be biased upward when the background rate is time-varying [Filimonov and Sornette, 2012], which our fixed seasonal mitigates but does not eliminate.

6.8 Relation to Rough Volatility

Nearly unstable Hawkes processes with heavy-tailed kernels converge, under rescaling, to rough fractional processes [Jaisson and Rosenbaum, 2016], which is the microstructural route to the rough-volatility models of Gatheral et al. [2018] and to rough Heston [El Euch and Rosenbaum, 2019]. In that limit a kernel tail exponent $1+\alpha$ corresponds to a Hurst parameter $H = \alpha - \frac{1}{2}$, so the $H \approx 0.1$ typically reported for volatility corresponds to a tail near 1.6.

Earlier drafts, working from the unestimated 1.36, reported our exponent as sitting closer to that rough-volatility value than to the Hawkes-kernel literature’s ≈ 1 . Estimation does not support that reading, but neither does it cleanly support the opposite one. The cross-kernel estimate, $\hat{\beta} = 1.033$ with a 95% interval of $[0.963, 1.104]$, rejects 1.6 and is consistent with 1.0; the rebuilt self-kernel estimate, 1.43–1.58, points the other way. Under the $H = \alpha - \frac{1}{2}$ mapping the two imply $H \approx 0.03$ and $H \approx 0.5$ respectively—the first rougher than the $H \approx 0.1$ usually reported, the second not rough at all. We conclude that our data do not currently locate the volatility process on this axis, and we decline to place it. **[pending: the exponent conditions of Jaisson and Rosenbaum [2016] on α must be checked before this mapping is asserted rather than noted; our $\hat{\alpha} \approx 0.03$ sits near the edge of the admissible range]**

The practical consequence for the benchmark set is unchanged and if anything strengthened: a fractional-kernel forecaster is a rival in the same power-law family as our backbone, our measured exponent puts us in the same corner of that family as the Hawkes literature, and the battery still does not contain one. **[pending: fractional/RFSV arm on the shared panel]**

This has a direct consequence for the paper’s benchmark set that we regard as a genuine gap: a fractional-kernel forecaster is a *rival in the same power-law family* as our backbone, and the battery does not contain one. It is cheap to add, since the direct power-law convolution machinery of Section 9 already constructs arbitrary $u^{-\beta}$ features. **[pending: fractional/RFSV arm on the shared panel]**

6.9 The Clock Organ

A Hawkes process of the form (19) is time-homogeneous. Intraday realized variance emphatically is not, and the calendar block is the single largest organ we measure (-0.0023 QLIKE, on the organ-ablation cache rather than the frozen battery). The backbone is therefore $\text{kernel} \times \text{clock}$ rather than kernel alone.

The clock does more than scale the kernel. Session-conditional estimation shows the target kernel *rotating* through the day—cosine similarity to the pooled kernel of $+0.83$ overnight against -0.55 at the open (**[pending: tab:hawkes _kernel]**)—so the geometry deforms at session transitions rather than merely breathing in amplitude.

We can now say how much. Stacking the six session kernels as a 6×12 matrix, the leading singular value carries only **43.8%** of the variance, and three components are needed for 84%. Pure amplitude motion on a fixed shape would put essentially all of it in the first component. The session variation of the *self*-kernel is therefore a genuine rotation, not a breathing amplitude—a distinction that matters because the exogenous operator, by contrast, does behave as the amplitude story predicts (Section 6.4). The two objects move differently through the day, and only the exogenous one has quasi-static geometry.

This is the hook that Section 8.7 pays off: the nonlinear structure that survives is anchored to exactly these transitions.

6.10 Summary: The Instrument

The backbone is a truncated power-law self-kernel with a clock organ, of which the base-5 HAR ladder is a coarse quadrature. The quadrature evidence is solid: it comes from the frozen forecast battery, it is monotone across six bases, and it does not depend on any kernel estimate. The kernel estimate is solid once correctly conditioned: the marginal self-kernel has integrated mass 0.913, no negative mass, and a power-law decay at $\hat{\beta} = 1.506 \pm 0.035$ (Section 6.6). What is not solid is any diagnostic computed on the self-kernel *net of* volatility proxies, which is what earlier drafts reported. Three consequences follow:

1. **The incumbent is quadrature-limited.** The base-5 ladder used as this paper’s shared benchmark is beaten by base-2 OLS by 0.00082 QLIKE at DM $t = -10.6$, with no change in information set, model class, or tuning. Section 7.1 accounts for what this does to every increment reported downstream.
2. **The natural basis is a mixture of decay exponents.** Direct power-law convolution features $K_\beta(t) = \sum_u u^{-\beta} y_{t-u}$, normalised, span the same structure as the boxcar ladder but are parameterised by decay exponent rather than window length—the natural coordinate if the kernel reading is correct, and freshness-robust in a way boxcars are not (a 3,125-wide rolling mean weights a 65-day-old bar exactly as heavily as yesterday’s).
3. **Two later negative results are predictions, not surprises.** Bounded additive-plus-pairwise nonlinearity and a low-rank exogenous response operator are what the quadratic and multivariate extensions of this backbone respectively predict, and the rank is measured at 4–6 of 41 channels. Both remain consistency arguments: we fit no self-exciting model, and a 30-minute aggregated variance is not the event stream such a model describes.

7 Where the Information Lives

With the backbone fixed, we measure what exogenous information adds to it. The measurement is made inside a single model class so that attribution is never confounded with model capacity: Section 8 varies the class with the information set held fixed, and this section does the reverse.

7.1 Benchmark Debt

Section 6.3 established that the shared incumbent—OLS on the base-5 HAR ladder plus calendar, QLIKE 0.13415—is beaten by the same estimator on a base-2 ladder, QLIKE 0.13332, at DM $t = -10.6$. That gap of 0.00082 involves no additional information, no change of model class, and no tuning. It is pure quadrature error in the benchmark.

We state the consequence before reporting any increment, because it applies to all of them. Every Δ QLIKE in this section and in Section 8 is measured against the base-5 incumbent, and therefore *includes* that 0.00082 of backbone misspecification as though it were exogenous or nonlinear content. For the smaller effects this is not a correction but a reclassification: the market-equal-weighted bucket’s apparent 0.00096 gain is, against a correctly specified linear backbone, on the order of the quadrature error itself.

The correct remedy is to re-run the bucket battery on the base-2 backbone so that the panel, the benchmark and the ladder convention are unified. **[pending: bucket battery on the b2 backbone; this is a correctness prerequisite, not a robustness extension, and until it lands every increment in Tables 7 and 9 should be read as an upper bound carrying 0.00082 of benchmark debt]**

Two mitigating observations. First, the debt is a constant additive offset shared by every arm, so *orderings* within a table are unaffected; it is levels and the significance of the smallest effects that move. Second, the largest effects in the paper—the joint exogenous gain of 0.00627 and the model-class results of Section 8—survive the correction comfortably, while the bottom half of Table 7 does not survive it cleanly.

7.2 Every Bucket Helps, and They Heavily Overlap

Table 7 reports the causally tuned ridge arm on each of the eight exogenous buckets, on the shared panel, against the shared incumbent. Ridge is the attribution instrument: it is the dense-shrinkage estimator that Section 8 shows to be the appropriate linear response to this regime, and holding it fixed while varying the information set isolates information from capacity.

Table 7: Bucket attribution. Causally tuned ridge on each exogenous bucket, shared panel ($n = 218,934$), identical rows for every cell. Δ is measured against the shared incumbent (OLS HAR base-5 + calendar, 0.13415) and carries the benchmark debt of Section 7.1. Δ_{own} is measured against the same estimator’s own no-exogenous baseline (0.13445), which is the correct within-class increment. Negative DM t favours the bucket arm.

Bucket	QLIKE	Δ vs. incumbent	Δ_{own}	DM t
all_features (joint)	0.12788	−0.00627	−0.00657	−19.9
moments	0.13085	−0.00331	−0.00360	−19.2
liquidity	0.13120	−0.00295	−0.00325	−13.4
implied_vol	0.13252	−0.00163	−0.00193	−11.3
market_vw	0.13291	−0.00125	−0.00154	−9.1
market_ew	0.13319	−0.00096	−0.00126	−7.2
vol_demand	0.13337	−0.00078	−0.00108	−4.1
sentiment	0.13347	−0.00068	−0.00098	−5.7
(no exogenous)	0.13445	+0.00030	—	+5.0

Two facts organise the table.

Every bucket improves on the backbone, decisively. All seven singleton buckets carry DM t between −4.1 and −19.2. Taken individually these are overwhelming; Section 7.5 explains why that is less impressive than it looks.

The buckets overlap almost completely. Summing the seven within-class increments gives 0.01364, against a joint increment of 0.00657: a ratio of 2.08. *The sum of the parts is roughly twice the whole.* The same computation on the causally tuned LightGBM arm gives 2.24, so the redundancy is a property of the information rather than of the linear estimator.

We note a trap here that the earlier drafts of this analysis fell into. If the overlap ratio is computed against the *shared incumbent* rather than each estimator’s own baseline, every bucket’s increment silently includes that estimator’s model-class premium, and the ratio inflates—to 3.98 for LightGBM, against its true 2.24. Overlap ratios must be computed within class.

The practical reading is that the eight buckets are eight noisy views of a small number of underlying states, not eight independent information sources. This is the same fact as the dense-but-weak finding of Section 6.1, observed one level up, and it is what the low-rank operator structure of Section 6.4 would predict.

7.3 Block Attribution

A Frisch–Waugh–Lovell block decomposition assigns each block its Type-III unique contribution—its gain net of everything else in the design.

Table 8: FWL block attribution of the linear base. Each entry is the block’s unique contribution net of all other blocks. **Panel note:** this decomposition was computed on the earlier base-5 exogenous cache (full fit QLIKE 0.12314) and is *not* on the frozen battery panel; it is reported here for its structure, not for level comparability with Table 7.

Block	Columns	Unique contribution
HAR backbone	6	−0.02208
Exogenous	359	−0.00449
Regime (HAR $\times$ {open, close})	—	−0.00225
<code>cumrv</code> (engineered)	1	−0.00122

The backbone accounts for 96% of explainable variance ($R^2 = 0.594$ of 0.617 for the full fit). Everything built on top of it is small. The per-feature comparison is the one the attribution literature would find striking: the single engineered path feature `cumrv` contributes −0.00122 on its own, while the 359-column raw exogenous block contributes −0.00449 in total, or 1.25×10^{-5} per column—a factor of

$$\frac{0.00122/1}{0.00449/359} \approx 97.5 \quad (24)$$

in favour of one mechanism-targeted measurement over piling on raw series.

We show the arithmetic because per-feature normalisation is easy to abuse. Two caveats belong with the number. First, dividing a block’s *unique* (post-FWL) contribution by its column count penalises internal redundancy twice: the exogenous block is heavily collinear, so its Type-III contribution is small precisely because its members substitute for one another, and then that small number is divided by the large count that caused the substitution. Second, `cumrv` was engineered *after* inspecting this panel, whereas the raw exogenous series were not; the comparison is between a searched-for feature and an unsearched block. The honest claim is directional: mechanism-targeted measurement is far more efficient per column than indiscriminate accumulation of price-derived series. The factor of 97.5 should not be quoted as an effect size.

Discrepancy note. An earlier internal memo records this ratio as “ $\sim 27\times$ ”. Recomputing from the block contributions above gives 97.5. The memo figure appears to be an error and should not be propagated.

7.4 The Model Confidence Set Over Feature Sets

Earlier drafts asserted that the 90% model confidence set [Hansen et al., 2011] over the nine feature sets is the singleton containing the joint bucket. **We withdraw that claim.** A model confidence set requires the arms’ per-bar loss series, which were not retained for the battery, so the assertion was never supported by a computed set; Section 11.4 reports the confidence sets we *can* compute, on the campaign tile, where they behave quite differently. **[pending: battery MCS over the nine feature sets once per-bar losses are persisted]**

Two interpretive cautions will survive the eventual computation. First, with $n = 218,934$ bar-level loss differentials an MCS has enormous power, and collapsing to a singleton is close to the default outcome whenever one arm is even slightly and systematically ahead; such a result would mean “the joint bucket is not statistically tied with any singleton bucket”, not “the joint bucket is the only useful feature set”—a distinction Table 7’s overlap ratio already makes on economic grounds. Second, the tile results run the other way: at $n = 2,189$ the MCS retains 123 of 161 arms and all 20 of the best. Sample size, not model quality, drives how discriminating a confidence set looks, and neither extreme should be reported without the other.

7.5 Multiplicity

Every DM statistic in Table 7 is computed against one shared comparator, and the full battery from which the table is drawn contains 98 arms, all compared to the same incumbent, with the best reported as the headline. No multiplicity adjustment is applied anywhere in this paper.

Section 11.1 now measures the cost. Across all 97 testable arms, 93 are significant at an unadjusted $p < 0.05$; Holm–Bonferroni leaves 91 and Benjamini–Hochberg leaves 93. Every bucket effect in Table 7 survives Holm, including `vol_demand` at $t = -4.1$, whose raw $p = 4.3 \times 10^{-5}$ becomes 4.2×10^{-3} after correction. Multiple testing is therefore not what threatens the bottom of this table—the benchmark debt of Section 7.1 is, and that threat is unaffected by any p -value adjustment. Of the two arms in the whole battery that do lose significance under Holm, one is the marginal `reclasso/sentiment` arm and the other a base-5 ladder cap variant; neither carries a claim in this paper.

7.6 Where the Attribution Points

The attribution results point outward rather than inward. The backbone carries almost all of the explainable variance; the exogenous block adds a real but small increment that is heavily redundant internally; and a single feature built to measure a specific intraday mechanism outperforms the raw block on a per-column basis by a wide margin. The implied next investment is not a larger model but mechanism-targeted data—auction imbalance, order flow, and dealer positioning aligned to the session transitions where Sections 6.9 and 8.7 both locate the structure.

8 Model Class: Does Extraction Require Nonlinearity?

This section holds the information set fixed and varies the estimator. All arms are causally tuned by the protocol of Section 9.1, so no arm’s hyperparameters ever see the losses they are scored on, and all sit on the shared panel.

8.1 The Comparison

Table 9: Model class at fixed information. Best tree (LightGBM) against best linear (ridge) on each bucket, causally tuned, shared panel ($n = 218,934$). Δ_{class} is tree minus linear on the same feature set. OOS- R^2 is raw-variance MSE relative to the shared incumbent; MZ- β is the Mincer–Zarnowitz slope, with 1.0 denoting an unbiased forecast.

Feature set	LightGBM	ridge	Δ_{class}	LGBM OOS- R^2	LGBM MZ- β
all_features	0.12604	0.12788	−0.00184	+0.059	0.892
moments	0.12816	0.13085	−0.00269	+0.021	0.882
liquidity	0.12840	0.13120	−0.00280	+0.058	0.892
market_vw	0.12963	0.13291	−0.00328	−0.010	0.862
vol_demand	0.12979	0.13337	−0.00358	−0.035	0.861
implied_vol	0.13001	0.13252	−0.00251	−0.034	0.840
sentiment	0.13038	0.13347	−0.00309	−0.063	0.867
market_ew	0.13039	0.13319	−0.00280	+0.003	0.867
baseline	0.13118	0.13445	−0.00327	−0.040	0.870
incumbent (OLS)		0.13415		—	0.929
OLS, base-2 ladder		0.13332			

The tree holds a modest edge on every one of the nine feature sets, ranging from 0.00184 to 0.00358, and Section 11.2 shows every one of those nine differences to be statistically resolved under an assumption-free bound. The consistency of the sign remains the most persuasive part of the evidence—an exact sign test over the nine sets gives $p = 0.0039$ —but the individual comparisons no longer rest on it alone. The magnitude is small.

The pairwise difference is tested, and it holds. Every DM statistic in the battery is computed against the shared incumbent, a third model, so $t = -15.4$ for LightGBM and $t = -19.9$ for ridge on **all_features** do not by themselves establish that the 0.00184 between them differs from zero. Section 11.2 supplies the missing test. Because the incumbent cancels from the pairwise differential, the pairwise standard error is bounded above by $se_{\text{tree}} + se_{\text{linear}}$ without any assumption about the dependence between arms, and the resulting bound resolves *all nine* matched comparisons in the tree’s favour after Holm adjustment (**[pending: tab:pairwise_bounds — per-pair bound table]**). The headline pair is the weakest of the nine, at $|t| \geq 2.19$, $p \leq 0.029$; the other eight range from $|t| \geq 3.73$ to $|t| \geq 5.91$.

Three caveats travel with that result. The reported statistics are bounds, so the true evidence is stronger by an unmeasured amount. The headline pair is only just resolved, which means the paper’s most-quoted number is its least secure. And a model confidence set over the 45 arms still requires their per-bar losses, which are not retained. **[pending: battery MCS once per-bar loss series are persisted]**

8.2 The Pure Nonlinearity Premium, Corrected

On the **baseline** feature set—HAR and calendar only, no exogenous information—the causally tuned LightGBM reaches 0.13118 against the incumbent’s 0.13415, a gap of 0.00297. This is the quantity earlier drafts reported as the pure nonlinearity premium, on the grounds that the two arms share an information set exactly.

Two corrections apply.

Backbone misspecification. The incumbent is quadrature-limited (Section 7.1). Measured against the base-2 OLS ladder at 0.13332, the premium is

$$0.13332 - 0.13118 = 0.00215, \tag{25}$$

so approximately 28% of the headline figure is HAR lag-grid quadrature error rather than nonlinearity. A tree on a log-spaced boxcar basis can interpolate between rungs; part of what it was being credited for is doing the linear model’s quadrature for it.

Protocol asymmetry. The comparison sets a *causally tuned* LightGBM against an *untuned* OLS. The matched linear arm on the same features—causally tuned ridge, 0.13445—is worse than the untuned incumbent. The premium is therefore measured against whichever linear specification happens to look best, which is not a clean isolate. We report it as a range: against the base-2 OLS ladder, 0.00215; against the causally tuned ridge on the same features, 0.00327. The lower figure is the conservative one and is the number we carry forward.

8.3 Tuning Is a Non-Lever

Three independent measurements agree that hyperparameters are not where the skill is.

1. **Causal tuning trails fixed defaults on the linear arms.** On the **baseline** feature set, causally tuned ridge (0.13445), elastic net (0.13471) and lasso (0.13475) are all *worse* than the fixed incumbent at 0.13415, at DM $t = +5.0$, $+7.5$ and $+7.0$ respectively. Selecting a penalty causally, on a leakage-clean validation tail, costs more in selection noise than it recovers in fit.
2. **An oracle over penalties has almost nothing to give.** A test-set oracle—selecting the penalty that minimises realised out-of-sample loss, an unattainable upper bound—improves the incumbent by at most 0.00015, an order of magnitude below the information effects.
3. **Selection loses to shrinkage at every layer tested.** Heavy ℓ_1 hurts; PCA and PLS truncation hurt; light shrinkage that prunes nothing important is the best linear response. This is the estimator-side restatement of dense-but-weak.

8.4 A Cautionary Result: Tuning on the Evaluation Panel

An earlier version of this study selected hyperparameters by minimising QLIKE on the full walk-forward backtest—the same panel and the same loss used for evaluation. We report the outcome because it is a useful calibration on the literature rather than a result about volatility.

Under that protocol the tuned tree ensembles reached QLIKE values around 0.11, roughly 0.02 better than any arm in the causally tuned battery, and the linear-versus-tree ordering *reversed* relative to the default configuration. Trial counts differed sharply by model (125 for XGBoost, 30 for LightGBM, 16 for random forest), so the arms were ranked partly by how many draws each had taken from the evaluation panel. The apparent effect, ≈ 0.03 QLIKE, is an order of magnitude larger than every genuine model-class effect in Table 9.

The introduction claims that published disagreements about whether nonlinear models help on this task are partly mechanical. This is the mechanism, measured on our own panel: a tuning protocol that touches the evaluation set can manufacture a model-class reversal several times larger than the effect under study. We regard the earlier numbers as uninterpretable and they appear nowhere else in this paper.

8.5 The Metrics Disagree

QLIKE is our primary ranker, on the standard grounds that it is robust to noise in the volatility proxy [Patton, 2011]. That choice is consequential here in a way that a single reported metric would hide.

Across the 45 causally tuned arms, the rank correlation between QLIKE and raw-variance OOS- R^2 is +0.152 ($p = 0.32$). Since lower QLIKE and higher R^2 both denote better forecasts, agreement would produce a *negative* correlation. The observed value is near zero, with the wrong sign, and is not distinguishable from no relationship at all: the two metrics do not rank this battery in a related way.

Concretely, **14 of the 42 arms that improve on the incumbent under QLIKE are worse than the incumbent under MSE**. These include the second-best arm overall (xgb / all_features, QLIKE 0.12641, OOS- $R^2 = -0.062$) and the best linear arm (ridge / all_features, QLIKE 0.12788, OOS- $R^2 = -0.058$). Only the single best arm, lgbm / all_features, improves on both.

Raw-variance MSE on this panel is dominated by a small number of extreme bars, most of them in March 2020, which is the standard argument for preferring QLIKE. We accept that argument but note that it is a *post hoc* defence of the metric under which our own ordering looks best, and that it should be settled with evidence rather than assertion. **[pending: MSE recomputed with the COVID window excluded, reported alongside the full-sample MSE, so that the reader can see whether the QLIKE/MSE divergence is a tail-domination artifact or a genuine disagreement about central performance]**

8.6 QLIKE Rewards Attenuation

Every arm in the battery is miscalibrated in the same direction: Mincer–Zarnowitz slopes run from 0.840 to 0.941, all below unity, indicating forecasts that are systematically too flat

relative to realised variance.

The pattern within that is the informative part. Across the 45 causally tuned arms, the rank correlation between QLIKE and MZ- β is **+0.741** ($p = 5.8 \times 10^{-9}$): *the arms that score best on QLIKE are the most attenuated*. The best arm, `lgbm / all_features`, has $\beta = 0.892$ against the incumbent’s 0.929; the second-best, `xgb / all_features`, has the worst calibration in the entire battery at $\beta = 0.840$.

This does not invalidate the ranking—QLIKE is a strictly consistent scoring rule and the ordering it induces is the one it is designed to induce—but it does change what the ranking means. QLIKE penalises under-prediction far more heavily than over-prediction, so a forecaster that shades toward the conditional mean of the loss rather than the conditional mean of the target is rewarded. A practitioner who cares about calibrated variance forecasts, rather than about QLIKE, would rank these arms differently, and the incumbent would move up. We report this as a finding about the loss function rather than about the models.

8.7 The Nonlinearity Saturates

The recoverable nonlinear structure has a ceiling, and the ceiling is low.

- Restricting the tree to additive-plus-pairwise structure anchored to the session clock recovers essentially all of the tree’s advantage; deeper interaction capacity does not pay.
- Deep sequence models and mixture-of-experts routing fail to improve on the saturated form.
- On the kernel dictionary of Section 6, trees fail as a *direct* forecaster, and fail decisively: per-bar DM on the campaign tile puts the extrapolation clamp at +0.077 for LightGBM ($t = 6.46$) and +0.102 for XGBoost ($t = 6.96$), the largest single effect in this paper (Section 11.3). As a *residual reader* the tree is worse than a smooth linear reader by ≈ 0.003 , but that difference is *not* statistically resolved ($t = 1.06$ and 1.11). We therefore claim the failure for the direct seat only. The nonlinearity that remains is smooth and diffuse rather than tree-shaped, which is consistent with—though not identified by—a second-order feedback.

Under a generic machine-learning framing this reads as an unexplained ceiling. Under the backbone of Section 6.4 it reads as a prediction. The quadratic Hawkes model of Blanc et al. [2017] makes the feedback a quadratic form in the past rather than a linear functional—generating clustering, fat tails and the Zumbach effect endogenously—and what it implies about extractable structure is exactly additive plus pairwise and no more. The saturation we measure is the saturation that model predicts.

We are careful about the strength of this claim. We have not fitted a quadratic Hawkes model, and the agreement is qualitative: a measured ceiling at pairwise order matching a model class whose feedback is of second order. It is consistency, not identification. **[pending: fitting the quadratic feedback form directly and testing it against the saturated tree would convert this from consistency to evidence]**

8.8 Summary

Within the dense-but-weak regime, nonlinearity buys a small, consistent, bounded amount—between 0.00184 and 0.00358 depending on feature set, with a pure premium of 0.00215 against a correctly specified linear backbone—and hyperparameters buy nothing. Set against the joint information effect of 0.00627, the ordering *information* > *model class* > *hyperparameters* holds on this panel by roughly a factor of three at each step. That ordering is the paper’s central practical claim, subject to the pairwise significance test that Section 8.1 flags as outstanding.

9 Algorithm Design

With the feature set frozen, the estimation *algorithm* becomes the design space. This section presents the measurement instrument the rest of the paper depends on—a causal hyperparameter-selection protocol—together with the exact every-bar estimators that make it computable, and one proposed nonparametric estimator whose ablations are decisive against its novel component.

9.1 Causal Hyperparameter Selection

Section 8.4 showed what happens when hyperparameters are selected on the evaluation panel. The protocol below is the replacement, and every arm in Sections 7 and 8 is estimated under it.

Every 250 solves, each arm re-selects its own hyperparameters from a leakage-clean forward split of its *current* training window (500 trading days, 24,000 bars):

1. a fit block, taking all but the last 150 bars of the window;
2. a 25-bar embargo, chosen to cover the longest moving average entering the features, so that no validation row shares a rolling window with a fit row;
3. a 125-bar validation tail, on which configurations are scored by mean squared error.

The winning configuration is then used for the next 250 forecasts and discarded. No forecast’s hyperparameters are ever selected using information from that forecast’s own future, and no forecast’s hyperparameters are selected on QLIKE, so the evaluation loss is never the selection loss.

Linear arms (ridge, recursive lasso, recursive elastic net) search log-scaled penalty grids and slide warm between tunings via the exact updates of Section 9.2, cold-reseeding at each re-selection. Tree arms search max-depth $\{3, 5, 8\} \times \text{colsample} \{0.25, 0.5\}$ with the learning rate pinned at the production value, fit grid cells at 300 trees and refit the winner every bar at 500. Model refit is every bar for all arms, so the refit-cadence asymmetry that confounded earlier comparisons is absent.

We regard this protocol, rather than any individual arm, as the section’s primary contribution: it converts “tuning does not help here” from an anecdote about one search into a claim with a defensible causal structure.

9.2 Exact Every-Bar Estimators

Per-bar refitting over 218,934 bars, crossed with 45 arms and a periodic hyperparameter search, is only feasible if the updates are exact and incremental.

Rank-1 rolling least squares. Adding and dropping one observation from a rolling-window least-squares fit are rank-1 modifications of the Gram matrix, handled by the Sherman–Morrison identity. This makes the every-bar OLS incumbent exact and computable at panel scale, and it is what allows the benchmark to be a genuinely per-bar refit rather than a cadence approximation.

Online elastic-net homotopy. The ℓ_1 analogue of the same idea. As the window slides, the elastic-net solution path is updated by homotopy continuation [Garrigues and El Ghaoui, 2008]: the active set and the signs of the active coefficients change at a finite number of break-points, between which the solution is piecewise linear in the perturbation. This delivers the exact recursive lasso and elastic-net arms rather than warm-started approximations, so that the “selection loses to shrinkage” finding of Section 8.3 is not confounded with optimisation error.

Block ridge. Where the design partitions naturally—backbone versus exogenous block—separate penalties per block are solved jointly, which is what allows the backbone to be lightly penalised while the wide exogenous block is shrunk hard.

9.3 Spectral k -Nearest Neighbours

The proposed estimator asks whether the dense-but-weak regime rewards analog retrieval: embed trailing residual trajectories with Laplacian eigenmaps [Belkin and Niyogi, 2003], retrieve past episodes that are near in the embedded geometry, and forecast by locally averaging their outcomes. The design has four axes: the residualiser (OLS-anchored or identity), the trajectory window W , the embedding dimension d (with $d = 0$ denoting passthrough, i.e. no embedding), and the neighbourhood size k .

9.3.1 The embedding hurts on every matched cell

Table 10: Embedding ablation. Each row is a matched pair differing only in whether the Laplacian-eigenmap step is applied ($d = 6$) or bypassed ($d = 0$). Shared panel, $n = 218,934$. Positive Δ means the embedding is worse. It is worse in all eight matched cells.

Window	Config	$d = 0$	$d = 6$	Δ
$W = 12$	OLS-anchored, $k = 100$	0.13219	0.13364	+0.00145
$W = 24$	OLS-anchored, $k = 100$	0.13218	0.13399	+0.00182
$W = 12$	OLS-anchored, $k = 25$	0.13405	0.13676	+0.00270
$W = 24$	OLS-anchored, $k = 25$	0.13397	0.13701	+0.00304
$W = 12$	identity, $k = 25$	0.14173	0.15993	+0.01820
$W = 12$	identity, $k = 100$	0.14236	0.16213	+0.01977
$W = 24$	identity, $k = 100$	0.14539	0.16874	+0.02336
$W = 24$	identity, $k = 25$	0.14475	0.16872	+0.02397

The manifold-embedding step—the novel component of the design—hurts on all eight matched configurations, by between 0.00145 and 0.02397, and the damage is an order of magnitude larger without the OLS anchor than with it.

9.3.2 The cause is supervision, not extension

Earlier drafts attributed the failure to out-of-sample Nyström extension error: new points cannot be placed on the learned manifold accurately enough for retrieval in the embedded space to beat retrieval in the ambient space. That explanation is intuitive, it was never tested, and it is wrong.

The eigenmap is defective on two axes at once, and they are separable. It is *nonlinear and defined on the observation set*, so a test bar has to be extended in—the extension story. It is also *built from X alone*, in a panel where the leading variance directions carry almost none of the predictable variation in y : the first principal component holds 96.2% of the design’s variance and 0.02% of the explained target, while principal components 6 through 20 hold 0.18% of the variance and 79.9% of the explained target. Adding a linear principal-component arm at the same dimension separates the two, because a PCA projection is an exact linear map that applies to a test point verbatim and has no extension step of any kind.

Table 11 holds the anchor, the candidate pool (48,579 training views), the evaluation rows (24,288, fixed by a rule that inspects no arm) and the neighbourhood grid constant, and varies only the retrieval coordinate.

Table 11: Retrieval-metric race, $W = 24$, $k = 100$. All arms share the HAR-12 OLS anchor, the pool, the evaluation rows and the neighbourhood size; only the coordinate in which neighbours are found differs. Δ and t are matched-row Diebold–Mariano against the ambient arm, Holm-corrected across the 24-cell grid. Positive Δ means the arm is worse. Every estimator here—anchor, operator, principal-component frame, eigenmap and all scaling constants—is fitted on training rows only. The displayed $k = 100$ is the ambient arm’s own out-of-sample optimum, so it is a selected column; we quote it because it is the incumbent coordinate’s best case, and the ordering of the six arms is identical at $k \in \{25, 4000, 8000\}$, so no conclusion here depends on that choice.

Coordinate	Type	d	QLIKE	Δ	t
residual path only	ambient, no exog	24	0.16347	−0.01476	−15.5
path + state	ambient	516	0.17823	—	—
operator triplets	supervised	6	0.17283	−0.00540	−8.9
operator triplets	supervised	12	0.17549	−0.00274	−5.1
operator scores	supervised	5	0.18399	+0.00575	+11.9
principal components	unsupervised, linear	6	0.18375	+0.00552	+12.8
Laplacian eigenmap	unsupervised, nonlinear	6	0.18274	+0.00450	+9.8
no retrieval (anchor)	—	—	0.18322		

Three readings follow, and $W = 12$ reproduces the ordering.

Extension error is not the mechanism. PCA at $d = 6$ cannot have extension error, and it still loses +0.00552—more than the eigenmap’s +0.00450. Whatever the eigenmap’s nonlinearity costs, it is not the source of the damage; the unsupervised objective is.

Supervision is the mechanism. The supervised coordinate at the *same* dimension $d = 6$ —the trailing-path score together with $\sigma_k u'_k Z v_k$ for the top five singular triplets of the train-only fitted exogenous operator—*beats* the full 516-dimensional ambient view by 0.00540 ($t = -8.9$), while the unsupervised compressions of that same dimension lose decisively. An 86-fold reduction in dimension improves the metric. The compression budget was never too small; it was spent in the wrong directions.

This result reverses on the data version, and the reversal is instructive. On the uncorrupted panel of Section 4 the same arm was a null (+0.00015, $t = +0.3$, Holm-adjusted $p = 1.0$), and we previously reported it as such. A distance weights every coordinate by its scale, so a channel standardised by a collapsed denominator dominates the metric outright; the supervised projection was spending its six dimensions on an artefact. Repairing the fill does not merely change the level here—it converts a null into the second-best arm in the table. The operator’s singular vectors have the incidental virtue of living on the design axes—channels and lags—so they are fixed a priori, finite, and require no out-of-sample extension at all.

The exogenous block is a net cost to a distance. Matching on the residual path alone, discarding all 492 exogenous ladder coordinates, beats the composed view by 0.01476 ($t = -15.5$). The same block that a fitted linear map profits from degrades a metric, because a distance weights every coordinate by its scale rather than by its relevance. This is the dense-but-weak fact in its sharpest form: with many weakly informative channels, *fitting* and *measuring similarity* are not the same operation, and only the first tolerates breadth.

The revised claim, then, is narrower than the paper previously made and better supported: unsupervised geometry fails in this regime, at any dimension tested and whether linear or nonlinear. Supervised geometry does not fail—it merely fails to *improve* on the ambient view, which is the honest ceiling on the idea.

9.3.3 What is load-bearing is not novel

The same table identifies what does the work.

- **The OLS anchor is load-bearing:** at $d = 0$, replacing the identity residualiser with the OLS anchor is worth between 0.00768 and 0.01321 across matched cells—five to nine times the size of any model-class effect in this paper.
- **Large-neighbourhood averaging is load-bearing, up to a point:** moving from $k = 25$ to $k = 100$ is worth ≈ 0.0018 in the anchored, unembedded cells. Per-bar inference on the campaign tile confirms the effect and locates its limit: a neighbourhood of 8,000 beats one of 200 by 0.0204 ($t = -6.59$), but 16,000 is 0.0029 *worse* than 8,000 ($t = 3.74$), both Holm-significant (Section 11.3). The optimum is a large but finite neighbourhood, so the correct statement is not that more averaging is always better. Heavy local averaging rather than sharp local structure is what the regime rewards—the nonparametric restatement of dense-but-weak.
- **The embedding is not,** as above.

The surviving estimator is therefore an OLS-anchored, unembedded, heavily averaged analog search. It is a reasonable method and it is not novel. We report this as the section’s finding rather than burying it: in the dense-but-weak regime nonparametric flexibility is a cost to be justified, not a default.

9.3.4 The headline cell is a selected minimum

The best kNN arm reaches QLIKE 0.13096 against the incumbent’s 0.13415, at DM $t = -17.6$. Three qualifications belong with that number, and earlier drafts of this paper carried none of them.

It is the minimum of 32 cells. The kNN family contributes 32 arms to the battery, of which 8 beat the incumbent. Quoting the minimum with an unadjusted DM statistic, on the same panel used to select it, is a winner’s-curse figure. Family-wise correction across the battery does not dissolve it—its p survives Holm across all 97 arms (Section 11.1)—but a p -value adjustment does not address selection: the issue is that the cell was chosen for being

the minimum, not that its individual test is too liberal. The honest presentations remain either a pre-registered cell or a confirmation on held-out data. The replication campaign of Section 12 now supplies the out-of-panel test, and it bounds the result rather than extending it: on the replication basis, with a pre-registered three-view design (trailing residual paths, forecast-error trajectories, and a Laplacian-eigenmap embedding at the same dimension), k NN corrections *fail to improve* a strong linear base (the pooled arm at $t = +0.98$ against a $+2.0$ gate after an overlap-leak repair that had inflated it to $+28$), and the eigenmap view is the worst of the three. The supervised-projection finding above should therefore be read as panel-specific until confirmed; the transferable statement is the one both panels agree on — estimated geometries do not beat fixed ones without supervision, and retrieval corrections do not survive on top of a well-shrunk dense base.

Its bucket is the one that survived. The winning cell sits on the `market_vw` bucket with an exogenous mask; sibling buckets failed as singular under the same configuration. Until that failure is diagnosed, the bucket ranking within the k NN family carries no weight, and the choice of `market_vw` cannot be read as an information finding.

It does not trail every parametric arm. At 0.13096 the winning k NN cell numerically beats **25 of the 45** causally tuned parametric arms, including the baseline LightGBM (0.13118), baseline XGBoost (0.13163), and all three baseline linear arms (0.13445, 0.13471, 0.13475). An earlier draft asserted that it trails every parametric arm; that is false against our own battery.

Applying the pairwise bound of Section 11.2 sharpens the picture in both directions. The k NN cell is decisively *worse* than the best arms on the informative buckets— $|t| \geq 6.96$ against `lgbm/all_features` and $|t| \geq 6.22$ against `ridge/all_features`—and decisively *better* than baseline ridge ($|t| \geq 14.49$). Against baseline LightGBM, where the raw gap is only 0.00022, the comparison is **inconclusive** ($|t| \geq 0.30$). So the defensible claim is narrower than either draft: the anchored analog search beats the linear arms on the backbone feature set and loses to the tree and linear arms on the informative buckets, with the tree-on-backbone comparison unresolved.

9.4 Summary

With features frozen, exactness and anchoring matter more than expressiveness. The causal-tuning protocol makes the paper’s central negative result about hyperparameters defensible; the rank-1 and homotopy updates make an every-bar benchmark computable at panel scale; and the analog-retrieval experiment resolves into a clean decomposition in which the anchor and the averaging pay and the manifold does not. The novel component of the proposed estimator is the component that fails, and we report it at the same altitude as the parts that work.

10 Stability Across Eras

Every attribution result in this paper is a full-sample aggregate over twenty years. That is the convention in this literature and it hides something important on this panel.

10.1 The exogenous block helps in seven eras of eight

Table 12: Exogenous designs against the backbone, by era, under the rolling protocol. Negative means the design beats the backbone. The stripped-down amplitude design is included because its behaviour in the failing era is the point—and because that behaviour turned out to be an artefact of the data defect of Section 4 rather than a property of the design.

Block	Period	backbone	ridge on 492	stripped-down
1	2007-02–2009-04	0.15054	−0.00400	−0.00271
2	2009-04–2011-06	0.10915	−0.00222	+0.00526
3	2011-06–2013-08	0.11269	−0.00231	−0.00105
4	2013-08–2015-09	0.13143	−0.00265	−0.00258
5	2015-09–2017-11	0.12048	−0.00469	−0.00323
6	2017-11–2020-01	0.15594	−0.00591	−0.00510
7	2020-01–2022-03	0.17033	−0.00771	−0.00977
8	2022-03–2024-04	0.13511	+ 0.03321	+0.00347

The exogenous block beats the backbone in seven consecutive eras—including through 2008 and 2020—by 0.0022 to 0.0077, and then fails once by +0.03321, an order of magnitude larger than any era’s gain. That single block erases seven eras of advantage: over the whole panel the raw exogenous design sits at 0.13617 against the backbone’s 0.13571, a difference of +0.00046 that is indistinguishable from zero.

A twenty-year aggregate therefore reports “exogenous information helps” or “exogenous information does not help” depending on where the window ends, and both statements are true of their own windows. We regard the aggregate alone as insufficient reporting, in this paper and in the literature it follows.

10.2 The failure is 27 bars, and it is our own

Concentration within the failing era, over its 27,376 bars:

Table 13: Share of the era’s total loss differential carried by its worst bars.

Worst	Bars	Share of damage
0.1%	27	47.2%
1.0%	273	90.5%
5.0%	1,368	122.4%
10.0%	2,737	138.0%

Shares above 100% mean the remaining bars have a *negative* differential: on the typical bar of the failing era the exogenous block still helps, and the median per-bar differential is +0.00069 against a mean of +0.03318. Twenty-seven bars out of twenty-seven thousand carry half of it.

All ten worst bars fall on 13–15 October 2023, and one channel, `voldemand_all_open_only`, carries 97.2% of the negative swing. Zero of forty-one channels exceeded three times their prior-month range, so this is not a market event. It is the imputation defect of Section 4: a series absent from the raw data, forward-filled, scaled by a collapsed interquartile range into coordinates in the hundreds, and fitted a coefficient.

Two things follow. The reported `vol_demand` bucket effect (Section 11.1) rests partly on that artifact and is suspended pending re-estimation on observed bars. And the practical reading of the exogenous block is not that it stopped working—by 2024 the stripped-down design is helping again—but that carrying it exposes a forecaster to rare collapses that require detection rather than tolerance.

11 Inference

Earlier drafts of this paper reported every improvement against one shared comparator, applied no multiplicity control, and left the central model-class difference untested. This section repairs all three. It also reports what the repairs cost: two arms lose significance, one claimed effect does not survive family-wise correction, and one that the paper leaned on turns out never to have been resolved at all.

11.1 Multiplicity Across the Battery

Ninety-seven arms are compared against the shared incumbent, and the paper quotes the minimum in two places. Table-wide correction is therefore mandatory, not optional.

Of the 97 tests, 93 are significant at an unadjusted $p < 0.05$. Applying Holm–Bonferroni across the full family leaves **91**; Benjamini–Hochberg at the same level leaves 93. The two arms that lose significance under Holm are `reclasso/sentiment` ($p = 0.036 \rightarrow 0.215$) and the base-5 ladder arm `ols_har b5 cap960` ($p = 0.037 \rightarrow 0.215$).

The conclusion is favourable to the paper and worth stating plainly: the information and model-class effects are not artifacts of multiple testing. A t of -19 against a family of 97 survives any correction one might apply, and even the weakest bucket effect we claim—`vol_demand` at $t = -4.1$ —clears Holm comfortably. What does not survive is the marginal sentiment arm, which we no longer claim.

11.2 The Pairwise Model-Class Test

The paper’s model-class conclusion compares two arms that were each tested only against a third. That is not the test the claim requires, and the per-bar loss series needed for a direct Diebold–Mariano test between them are not retained. A bound is available without them.

For two arms A and B scored against a common incumbent, the incumbent cancels from the pairwise differential:

$$d_A - d_B = (L_A - L_{\text{inc}}) - (L_B - L_{\text{inc}}) = L_A - L_B, \quad (26)$$

so the quantity we want is exactly the difference of two reported means. Its variance is

$$\text{Var}(\overline{L_A - L_B}) = se_A^2 + se_B^2 - 2\rho se_A se_B, \quad (27)$$

with se_A , se_B recoverable from each arm’s reported mean differential and HAC t statistic. Taking the worst case $\rho = -1$ gives $se \leq se_A + se_B$, hence a *lower bound* on $|t|$ and an *upper bound* on p that assume nothing whatever about the dependence between the two arms. If the bound clears the critical value the difference is established; if it does not, the comparison is inconclusive rather than negative.

Table 14: Pairwise model-class inference on the frozen battery, from the reported HAC standard errors alone. For arms sharing an incumbent the pairwise differential is exactly $L_A - L_B$, whose standard error is bounded above by $se_A + se_B$ regardless of the correlation between the two arms’ differentials; the resulting $|t|$ is therefore a *lower bound* and the p an *upper bound*. A tighter bound assuming the two arms’ improvements co-move ($\rho \geq 0$) is reported in the text; it does not change any verdict. Holm adjustment is applied across the nine matched comparisons. Negative t favours the tree.

Feature set	ΔQLIKE	$ t \geq$ (free)	$p \leq$ (free)	$p_{\text{Holm}} \leq$	Verdict
all_features	-0.00184	2.19	0.029	0.029	tree better
moments	-0.00269	3.82	1.4×10^{-4}	4.1×10^{-4}	tree better
liquidity	-0.00281	4.59	4.5×10^{-6}	2.3×10^{-5}	tree better
market_vw	-0.00327	5.91	3.4×10^{-9}	3.1×10^{-8}	tree better
vol_demand	-0.00358	5.71	1.1×10^{-8}	9.1×10^{-8}	tree better
implied_vol	-0.00252	3.73	1.9×10^{-4}	4.1×10^{-4}	tree better
sentiment	-0.00310	5.27	1.4×10^{-7}	8.3×10^{-7}	tree better
market_ew	-0.00281	4.17	3.0×10^{-5}	1.2×10^{-4}	tree better
baseline	-0.00327	5.54	3.0×10^{-8}	2.1×10^{-7}	tree better

The model-class ordering is established. All nine matched comparisons resolve in the tree’s favour after Holm adjustment, under a bound that makes no assumption about the dependence between arms. The headline pair—best tree 0.12604 against best linear 0.12788—is the weakest of the nine, at $|t| \geq 2.19$ and $p \leq 0.029$. Under the tighter bound that holds when the two arms’ improvements co-move ($\rho \geq 0$), the same pair gives $|t| \geq 3.00$, $p \leq 0.0027$; we quote the assumption-free figure.

Two qualifications. First, these are bounds: the true statistics are larger in magnitude, so the evidence is stronger than the table shows, but by an unmeasured amount. Second, the headline pair is only just resolved. A referee would be right to note that the paper’s most-quoted number, 0.00184, is the least secure of the nine, and that the case for the model-class ordering rests more on its consistency across feature sets than on any single comparison.

A distribution-free check. The tree beats the linear arm on all nine feature sets. Under a null of no model-class effect, the exact binomial sign test gives $p = 0.0039$. The nine feature sets are nested and share a panel, so this is anti-conservative and we report it as a consistency check rather than as an independent test.

11.3 Per-Bar Inference on the Campaign Tile

Several claims in this paper are inherited from campaign experiments whose per-bar predictions *are* retained, on a common tile of $n = 2,189$ bars with identical targets across 161 arms. That tile permits genuine Diebold–Mariano tests and genuine model confidence sets rather than bounds.

Panel discipline. The tile is not the frozen battery. Its levels are not comparable to any level elsewhere in this paper, and its sample is two orders of magnitude smaller, so it has far less power. What it can settle is whether each *claim* survives per-bar inference at all. We report it separately for that reason and never mix its numbers with battery numbers.

Table 15: Per-bar Diebold–Mariano tests on the campaign tile ($n = 2,189$ bars, 161 arms, identical targets). HAC (Newey–West) standard errors with the Harvey–Leybourne–Newbold small-sample correction; Holm adjustment across the whole contrast family. Negative t favours arm A. **Panel note:** this is a campaign tile, not the 218,934-bar frozen battery; no level here is comparable to a battery level, and what the tile tests is whether each claim survives per-bar inference at all.

Contrast (A vs B)	ΔQLIKE	t	p_{Holm}	Verdict
trees direct: xgb vs incumbent	+0.10163	6.96	8.7×10^{-11}	B better
local k=8000 vs k=200	-0.02041	-6.59	9.8×10^{-10}	A better
trees direct: lgbm vs incumbent	+0.07684	6.46	2.3×10^{-9}	B better
cap256 vs cap32	-0.00735	-4.16	5.2×10^{-4}	A better
cap256 vs uncapped	-0.00774	-3.83	0.002	A better
local k=16000 vs k=8000	+0.00294	3.74	0.003	B better
cap256 vs cap128	-0.00264	-3.50	0.006	A better
enet mid vs ridge	+0.00523	3.15	0.020	B better
rank vs divide	+0.00718	2.73	0.070	unresolved
enet heavy vs ridge	+0.00573	2.20	0.277	unresolved
Fourier clock vs session gates	-0.00367	-1.64	0.908	unresolved
PCovR-50 vs PCovR-25	-0.00213	-1.56	0.954	unresolved
trees residual: xgb vs ridge cadence	+0.00321	1.11	1.000	unresolved
trees residual: lgbm vs ridge cadence	+0.00308	1.06	1.000	unresolved
PLS reg vs PCovR	+0.00054	0.68	1.000	unresolved
enet light vs ridge	-0.00012	-0.65	1.000	unresolved
tie: rawc stack vs champion	+0.00063	0.30	1.000	unresolved
tie: prune32+corr vs champion	+0.00014	0.06	1.000	unresolved
tie: legchamp+clock vs champion	+0.00001	0.00	1.000	unresolved

Four claims are confirmed at Holm-adjusted significance across the nineteen-contrast family.

- **Trees fail as direct forecasters on the kernel dictionary**, and fail badly: XGBoost +0.102 ($t = 6.96$) and LightGBM +0.077 ($t = 6.46$) against the incumbent. The extrapolation clamp reported in Section 8.7 is real and is the largest single effect in the paper.
- **Heavy local averaging pays**, decisively: $k = 8000$ beats $k = 200$ by 0.0204 ($t = -6.59$).
- **Kernel truncation is a genuine interior optimum**. Cap-256 beats cap-32 (-0.0074), beats no truncation at all (-0.0077), and beats cap-128 (-0.0026), all Holm-significant. Truncating the kernel is not a computational convenience; removing it costs real accuracy.
- **Selection that bites hurts**. Elastic net at $\alpha = 3 \times 10^{-4}$ is +0.0052 worse than ridge ($t = 3.15$), while light ℓ_1 is indistinguishable from ridge. This is the sharpest available form of “shrink, never select”.

Two results refine claims made elsewhere in the paper.

- **Averaging has an interior optimum too**. Moving from $k = 8000$ to $k = 16000$ is +0.0029 *worse* ($t = 3.74$, Holm-significant). The correct statement is not “more averaging is better” but that the optimum sits at a large but finite neighbourhood; Section 9.3 is amended accordingly.
- **The parsimony ties are ties**. The three compressed representations are statistically indistinguishable from the wide champion: $p = 0.998$, 0.951 and 0.763 respectively. A tie is not evidence of equality, but with these p -values the burden plainly lies with anyone claiming the wide model does something the compressed ones do not.

11.3.1 Three claims that do not survive

We report these at the same altitude as the confirmations.

1. **Trees as residual readers are not shown to fail**. LightGBM +0.0031 ($t = 1.06$, $p = 0.29$) and XGBoost +0.0032 ($t = 1.11$, $p = 0.27$) against the smooth linear reader. The sign is as claimed and the point estimates match earlier reports, but neither is distinguishable from zero. The claim that “trees fail in both seats” holds for the direct seat only.
2. **Fourier clock modulation does not beat session gates**. -0.0037 with $t = -1.64$, $p = 0.10$ unadjusted and 0.91 after Holm. An internal note recorded gates as losing on both tiles; on this tile that is a point estimate, not a result.
3. **Rank versus divide is not resolved**. +0.0072 with $t = 2.73$, $p = 0.0064$ unadjusted but 0.071 after Holm across the contrast family. An earlier note recorded $p = 0.001$ for this comparison; under HAC standard errors with a small-sample correction and family-wise adjustment it falls short at the 5% level.

11.4 Model Confidence Sets

Table 16: Hansen–Lunde–Nason model confidence sets on the campaign tile ($\alpha = 0.1$, $B = 5,000$ stationary-bootstrap replications, $T_{\max}$ statistic). “Kept” counts arms that cannot be excluded from the set of best models. Survivor counts for the top-twenty family are unchanged when the bootstrap block length is varied from half to four times its automatic value.

Family	Question	Arms	Kept	Eliminated
penalty	Penalty: shrinkage vs selection	4	2	pbe_enet_0.0003, pbe_enet_
trees_vs_linear	Model class on the kernel dictionary	6	4	pbt_xgb_cad, pbt_lgbm_cad
local_k	Analog neighbourhood size	6	3	i_loc_k200, incumbent, i_lo
cap_ladder	Kernel truncation depth	6	6	—
dim_reduction	Dimension reduction	8	8	—
top20	The twenty best arms	20	20	—
all	Every tile arm	161	123	(see text)

The MCS is a joint test and correspondingly less powerful than the targeted contrasts above, which shows in the results.

The discriminating comparisons discriminate. On the penalty family the MCS retains ridge and light elastic net and eliminates both heavier elastic nets ($p = 0.009, 0.023$). On the model-class family it retains ridge, both residual-reading trees and the incumbent, and eliminates both direct-forecasting trees ($p < 0.001$). On the neighbourhood family it retains $k \in \{1000, 4000, 8000\}$ and eliminates $k = 200, k = 16000$ and the incumbent.

The top of the tile is a wide tie. All twenty of the best arms survive, and the result is unchanged when the bootstrap block length is varied from half to four times its automatic value. Over the full set of 161 arms the MCS retains 123. With $n = 2,189$ this is a statement about power, not about the models: the tile cannot discriminate among the great majority of arms tried on it, and any ranking within that set—including the ordering of the top three—is not supported by the data.

Consequence for the campaign. The practice of reading a tile leaderboard as a ranking is not defensible at this sample size. Differences at the top of the tile are real only if they replicate at scale, which is the argument for the frozen-battery discipline adopted for the rest of this paper.

11.5 Auxiliary Tests

Ladder monotonicity. The quadrature prediction of Section 6.3 is that QLIKE increases monotonically with the ladder base. Across the six bases the rank correlation is exactly +1.000 ($p < 10^{-4}$): the ordering is perfect, with no inversions.

Metric disagreement. The rank correlation between QLIKE and raw-variance OOS- R^2 across the 45 causally tuned arms is +0.152 with $p = 0.32$. Agreement between the two metrics would produce a *negative* correlation; what we observe is indistinguishable from no relationship at all. The two metrics do not rank this battery in a related way.

Calibration. The rank correlation between QLIKE and the Mincer–Zarnowitz slope is +0.741 with $p = 5.8 \times 10^{-9}$. Arms that score better on QLIKE are more attenuated, and the association is overwhelmingly significant. Section 8.6 discusses what this implies for reading a QLIKE ranking.

11.6 What Remains Unresolved

- A full model confidence set over the 45 causally tuned battery arms still requires their per-bar losses, which are not retained. The pairwise bounds of Section 11.2 settle the model-class question but do not produce a confidence set over the battery. **[pending: re-run the battery with per-bar loss series persisted, then a battery MCS supersedes both the bounds and the tile MCS]**
- The spectral- k NN headline remains a minimum over 32 cells selected on the evaluation panel. Bounds against the parametric arms (Section 9.3.4) do not address that selection.
- Everything in this section is conditional on the reported HAC standard errors being computed as documented. The bounds inherit whatever bandwidth convention produced the battery’s t statistics.

12 An Independent Replication on the State Side

Everything to this point was estimated on one panel and one basis, by design (Section 13 is the cost accounting of that commitment). This section reports a replication campaign run on a second construction — a frozen state-space basis and its dynamics — with different machinery at every step. The underlying bar series is the same (the full 242,934-bar panel, of which the preceding sections evaluate the post-warm-up 218,934); the construction is not: a 543-column transform set replaces the raw channels, the benchmark is that panel’s own per-bar HAR composition, and QLIKE levels are therefore NOT comparable across the two conventions and are never compared. Diebold–Mariano statistics in this section are signed so that positive favours the named candidate. Numbers here are quoted only in this section and are never merged into the preceding tables; the panel note of the one-panel commitment stands. Every claim below was pre-registered with a recorded expectation before its run, and every surviving claim passed a closing Romano–Wolf step-down over its own family (2,000 draws, 1,008-bar joint blocks); the campaign retired sixteen further pre-registered arms at their gates. Where an expectation is reported as wrong, the direction of the error is part of the record.

12.1 The frozen frame and its stable objects

Project the ~ 133 base features onto the eigenvectors of their first-window (2005–06) correlation and freeze that frame permanently. Three geometric objects built on this frame replicate out of sample: a factor-pair interaction map (split-half correlation $+0.62$ against a null of 0.00 ± 0.11), a daily rotor (92% curl fraction, 35-day period, phase forecastable roughly one month ahead), and an intraday circulation (split-half $+0.992$, kill-tested against moving-average arithmetic). A dependence-adjusted Marchenko–Pastur bound keeps only 5 of 106 directions, yet wider blocks carry real signal: mid-spectrum directions are *gauge-degenerate* — unpinnable as labelled axes while their span is stable. This resolves a puzzle from the primary basis (why representative choice never matters) and predicts the central robustness result below.

12.2 The lead–lag field

Let $G(t)$ be the frame scores and D the antisymmetric part of their lag-one-bar cross-correlation (trailing 504 days, quarterly re-estimation). The columns of $\hat{G}_j(t) = \sum_i D_{ij} G_i(t-1)$ — one-bar lead-lag “arrows” — are genuine feature content: a raw-lag control (the same columns without D -weighting) fails where the D -weighted block passes.

Width. The frame width was an inherited convention (twenty factors). Varying it end to end gives the campaign’s cleanest structural result: forecast improvement is *flat on a plateau* of widths ~ 30 – 80 and falls off a cliff on both sides (at width 106 — every live direction — the block fails its gate outright). The arrows live in a finite-rank sub-span roughly twice the conventional frame; production runs width 40 as a representative of the plateau, not as a tuned optimum. Mechanically, the widening gain is carried by the *new directions’ own next-bar motion*, not by new information about the old carriers; doubling the D estimation window rescues nothing (the tail is empty, not noisy); and the new arrows replicate split-half at $+0.838$ (old block $+0.907$) — stable field, unpinnable labels, as Section 12.1 predicts.

Horizon. The block’s value is fast and strictly horizon-dependent, and the width increment is faster still:

Table 17: The transmission block by horizon (Diebold–Mariano t against the same-engine twin; all surviving entries FWER-controlled in the closing step-down). The width-40 increment over width-20 exists at the one-bar horizon only.

horizon (bars)	1	4	8–16	end-of-day
width-20 block vs. twin	+9.9	+5.0	+2.6 / +5.9	+1.4 (fail)
width-40 vs. width-20	+2.3 (per-bar +4.6)	+0.1	+0.4 / +1.6	—

The one apparent anomaly is itself lawful: at sixteen bars the block passes only under a per-horizon shrinkage scaling ($\alpha \propto H$), sixteen times the penalty of its one-bar seat — dense-weak content pays at long horizons only when the solver is forced to be humble, the same law the primary basis found for its own feature classes.

12.3 The second moment: predictability persists, complexity died

The forecast-error second moment (the smear term in the QLIKE reconstruction) is forecastable: daily log-amplitude carries an out-of-sample R^2 near 0.5 from its own trailing means. Two campaigns of dictionary expansion establish what that forecastability is worth. First, ridge over a union dictionary (ten named amplitude functionals plus the full daily panel snapshot, generalized-cross-validated shrinkage, causal per-refit standardization and Duan factors) *beats* the named columns ($t = +3.12$, FWER $p = 0.006$) — the channel was span-deficient, not sparse, and the deficiency is nameable: calendar structure (day-of-week, half-day sessions) and slow quarticity, bipower, and liquidity aggregates, with the named ten carrying under 8% of the fitted coefficient mass. Second, that surplus is *era-bounded*: decomposed by period, the union’s edge decays monotonically after 2015 and is dead after 2022, while the amplitude’s own-means R^2 is undiminished (2020 posts the sample’s best year). The second moment did not become unpredictable in the zero-days-to-expiry era; it became *simple* — means-sufficient — and every dictionary expansion beyond the means (panel span, quarticity, regime labels, event decay, a sequence model’s seat) died on the modern regime. The single exception obeys the obvious mechanism: the calendar fragment, stationary by construction, survives on 2022–26 ($t = +2.31$ there; FWER $p = 0.007$ in the closing step-down) and enters the production smear. The conditional smear also fails to transfer to the end-of-day horizon (session integration averages the vol-of-vol away) — the trading-horizon deliverable keeps a trailing constant, now by measurement rather than by omission.

12.4 Horizons and the deliverable

The dense edge is fast in horizon (composite-over-backbone t : +5.4 at two hours, +2.8 at four, +1.2 at eight), and at the end-of-day horizon it concentrates at the open: pooled training across horizons *hides* the open-decision edge (−0.2), while dedicated-horizon training recovers it (+3.2 on the 10:00 decision slice). The per-horizon production stack that falls out of the campaign: at one bar, the full 719-column model (679 primary columns plus the width-40 transmission block) refit per bar; at intermediate horizons, the width-20 block with $\alpha \propto H$ scaling; at end-of-day, neither — the plain 679 with a constant smear. Composed with its conditional smear — ten amplitude functionals plus the calendar block certified below — the one-bar deliverable’s QLIKE improved from 0.12579 to 0.12299 over the campaign ($\approx -7.3\%$ against that panel’s HAR baseline), with the final step — the measured width and the calendar smear — contributed entirely by arms that a family-wise error correction survived.

12.5 What this replication is evidence for

The campaign closes with a seven-for-seven family sweep — the study’s first with no casualties — against sixteen pre-registered arms retired at their gates along the way. Three of the primary basis’s laws re-derive on the state side with different machinery: density with generous uniform shrinkage beats selection (the width plateau; the $\alpha \propto H$ revival); the edge is fast and freshness-loving in the first moment (per-bar transmission entry, horizon-locality

of width) while the second moment obeys neither law (rolling windows and faster refits fail there); and clever structure survives exactly where a stable geometric object backs it (arrows replicate; the phase claim, the flow-state features, and the lagged-quadratic terms do not). The state side also contributes one law the primary basis could not see: regime change manifested not as decay of predictability but as *collapse of complexity* — a channel whose extra dictionary content dies while its first-order structure strengthens.

13 Withdrawn Claims

This paper went through three drafts whose central claim we subsequently withdrew. We publish the record rather than the final draft alone, for two reasons. The claims were individually plausible, several were statistically significant at conventional levels, and one survived long enough to have the paper rewritten around it—so the interesting content is not that they were wrong but *how* they passed. And a reader is entitled to know the denominator: fifteen withdrawn claims is the evidence that what remains was tested rather than merely found.

13.1 The Ledger

13.2 Five Causes, Not Fifteen

Every entry in Table 18 traces to one of five failures. None is a modelling error, which is why we now guard against these five specifically rather than against wrong answers in general.

One factor at a time. The stripped-down design beat the elaborate one on the uncorrected panel and lost on the corrected one: a design $\times$ data-version interaction, discovered by accident after two rewrites. The same error then repeated *inside* the data fix, where median substitution alone looked harmful, masked estimation alone looked inert, the IQR floor alone looked inert, and all of the value was in the interaction. Three attempts failed for one reason. We now run a factorial with the data version as an explicit factor.

Proxy metrics never validated. Scaled magnitude in rolling-IQR units stood in for forecast harm, and was wrong in both directions: it called `sumbipow_ewstock` at 84.6 IQRs a pathology when the composed fix takes it to 17.7, and it called an `effspread` regression at 932 a defect whose QLIKE cost was never measured. No claim denominated in IQRs is a claim about forecasting. The same error recurred a third time in the scale- collapse detector, which fired on a region the repaired pipeline neutralises by design.

Selection contaminating the estimate. The MIDAS shape parameter was chosen against the reported out-of-sample loss, worth +0.0033; an exponent sweep reported its argmin as achievable; penalties were inherited across a dataset change. Selection now runs on a region strictly preceding evaluation, and oracle arms are reported but labelled.

Table 18: Claims made and withdrawn during this study. None is reinstated anywhere in this paper.

claim	why it was withdrawn
the effective kernel is negative, $n = -0.038$	wrong estimand and wrong basis
the kernel estimate is not reproducible	rebuilt cache reproduces at $r = +1.0000$
the trial space is a manifold, not a subspace	a width argument prices uniform approximation over a family; a regression fits one point
one nonlinear degree of freedom beats twelve linear	oracle selection artefact, $+0.0033$
a manifold-drawn linear space cannot reach the nonlinear fit	arithmetically guaranteed reachable
QLIKE rewards attenuated forecasts	two attenuation operators disagree; inconclusive
the kernel ceiling is 0.26337	an estimator ceiling, beaten at 0.26181
directed shrinkage makes 41 weak channels usable	tail-risk control, then a data artefact
the stripped-down model beats the elaborate one	reversed on the corrected panel
the raw parameterisation is actively harmful	it faithfully carried a corrupted feature
the exponent’s usable range is $[-1, -0.5]$	flat over $[0.2, 3.0]$
23 of 41 channels show the volatility-demand pathology	the detector conflated collapsed scale with fat tails
the honest indicator is the whole of the fix	it changes no feature value
the median substitution is actively harmful	harmful alone, correct as half of a pair
three supervised lag directions beat all twelve rungs	the identical procedure on the eight estimation windows nobody chose puts all eight behind the ladder; the published window sits 13.5 sd outside that distribution

Numbers quoted across incompatible footings. QLIKE 0.13280 on the uncorrected panel was tabled against 0.13177 on the corrected one without labels. Every number in this paper now carries its data version.

An arbitrary nuisance choice fixed at one value and never varied. The lag-compression result estimated its projection on the first 24,000 rows of the panel. Nothing recommended those rows over any other 24,000; they were simply first. Every diagnostic we ran on that arm was internal to it—refit cadences, seed variants, principal angles, a memory-half-life sweep—and each was informative about *how* the projection behaved without ever asking whether a different window would have produced the same verdict. It would not: repeating the identical procedure on the eight windows nobody chose puts all eight behind the full ladder in a tight distribution (mean $+0.00022$, sd 0.00016), with the published window 13.5 standard deviations outside it. The window was a nuisance parameter, and a nuisance parameter examined at one value is not examined. We now sweep such choices across their plausible range and read the reported number against that distribution,

in the spirit of specification-curve and multiverse analysis; where a sweep is infeasible, the arbitrariness is stated rather than left implicit.

13.3 What This Implies About the Rest of the Literature

We make one observation and no accusation. Four of the five causes above are invisible in a finished paper. A reader cannot tell from a results table whether a shape parameter was selected on validation data or on the reported test set; whether a reported ablation varied one factor while a second moved underneath it; whether the panel a model was fitted on contained a channel that had stopped publishing; or whether an arbitrary window, seed, or split was the only one ever tried. Our own answers to those four questions were wrong, and we had the code.

14 Limitations and Threats to Validity

The paper’s third methodological commitment is that negative and incomplete results are reported at the same altitude as positive ones. This section discharges that commitment for the items that a reader should weigh against every number above. We separate threats that could change a conclusion from gaps that merely widen a confidence interval.

14.1 Threats That Could Change a Conclusion

Benchmark debt. All increments are measured against a base-5 OLS incumbent that a base-2 OLS beats by 0.00082 at DM $t = -10.6$ (Section 7.1). Orderings within tables are unaffected; levels and the significance of the smallest effects are not. The bottom four rows of Table 7 have increments of the same order as the debt and should not be treated as established until the battery is re-run on the base-2 backbone.

Target winsorization (measured: it is costing accuracy). Stage 3 clips each series to rolling 5th and 95th percentile bounds, modifying 10% of observations by construction, and on a heavy-tailed variance target the clipped mass is disproportionately the volatility spikes a forecast exists to anticipate. We have now run the ablation (**[pending: tab:kernel_experiments — ablation table not in this draft]**). On a rebuilt panel with a single 60/40 split, relaxing the clip to 1/99 improves raw-scale QLIKE from 0.2648 to 0.2449, and removing it entirely improves it further to 0.2411. The kernel diagnostics barely move.

The direction is unambiguous and the magnitude is large relative to everything else in this paper: winsorizing the target costs roughly 0.024 QLIKE, an order of magnitude more than the model-class effect the paper spends a section establishing. Two caveats keep this from being a headline. The split is a single 60/40 cut with a full-sample ridge probe, not the walk-forward harness, so the levels are not comparable to the battery’s; and the ablation runs on the rebuilt panel of Section 6.6, not the frozen one. **[pending: re-run the winsorization ablation inside the walk-forward harness on the frozen panel; on present evidence the 5/95 clip should be expected to disappear from the pipeline]**

The diurnal divisor is itself a forecast (measured). Stage 1 divides by a *rolling* 20-observation same-slot mean rather than a fixed seasonal profile (Section 5.2.1). That is a slowly adapting level estimate, and it is multiplied back in at evaluation time, so every model in the paper inherits a common predictive component.

We have now measured how large that component is. Used *alone* as a forecast—no model, no features, just the rolling same-slot baseline—it reaches raw-scale QLIKE 0.558 on a held-out 40% of the panel, against 1.320 for an unconditional mean and 0.241 for the best full model in the same setup. On that scale the divisor closes roughly 71% of the distance between an unconditional forecast and a fitted one. Replacing it with 48 slot dummies inside the full model costs 0.03 to 0.06 QLIKE ([\[pending: tab:kernel_experiments\]](#)), so it is not merely a normalisation convenience: it is doing forecasting work that the reported model differences are then measured on top of.

The implication for reading this paper is that our effect sizes are increments over an already-strong shared component, and comparisons against a naive benchmark that lacks it will overstate every model’s skill.

This divisor was also a suspect for the archived kernel’s sign structure. Section 6.6 exonerates it: the cause is the exogenous conditioning set, and the marginal kernel is well behaved with the divisor still in place.

Pairwise significance (resolved, with a caveat). The model-class difference is now tested directly, by a bound that requires no assumption about the dependence between arms and that resolves all nine matched comparisons after Holm adjustment (Section 11.2). The residual caveat is that the headline pair is the weakest of the nine at $|t| \geq 2.19$, and that a model confidence set over the 45 battery arms is still unavailable because their per-bar losses were not retained.

Multiplicity (resolved). All 97 testable arms are now adjusted: Holm leaves 91 of the 93 raw-significant arms and Benjamini–Hochberg leaves 93 (Section 11.1). No claim in this paper depends on either of the two arms that lose significance. Adjustment does not, however, address the selection of the spectral- k NN headline as the minimum of 32 cells: that is a different problem and it remains open.

Claims withdrawn after testing. Three findings inherited from campaign notes did not survive per-bar inference and are no longer asserted: that trees fail as residual readers ($t = 1.06$ and 1.11 , unresolved), that Fourier clock modulation beats session gates ($p = 0.10$ unadjusted), and that the rank marginal adjustment is worse than dividing ($p = 0.071$ after Holm, against $p = 0.001$ in the original note). A fourth, the singleton model confidence set over feature sets, was asserted without ever having been computed and is withdrawn outright (Section 7.4). We record these because the pattern matters: each was a directionally correct point estimate that had been promoted to a result.

14.2 Gaps and Inconsistencies

Feature accounting does not reconcile. The paper describes 41 exogenous channels, a 359-column exogenous block, and a 529-column design matrix. Under the stated expansion rule of $|\mathcal{L}| = 6$ rolling means per channel, $41 \times 6 = 246 \neq 359$, and $529 - 359 = 170$ is not accounted for by six backbone columns plus calendar. **[pending: an explicit channels $\rightarrow$ lag-expansion $\rightarrow$ design-columns table, per bucket; the three counts in circulation cannot all be right]**

Two training-window conventions. The preprocessing and backtest description specifies a 500-observation rolling window; the causally tuned battery uses 24,000 bars. Only the latter produced the numbers reported here. The former should be removed or explicitly scoped to the superseded experiments.

Panel heterogeneity in superseded tables. Earlier subgroup analyses compared rows with materially different N (as low as 138,279 against 222,058) and differenced them anyway. Buckets whose coverage begins late—social sentiment and options order flow—span a different calendar era from the baseline they were differenced against, so those increments confounded information with regime. All numbers in this version come from the frozen battery, on which every arm shares $n = 218,934$ identical rows; we note the issue because it explains why earlier internal numbers disagree with these.

The kernel was measuring the wrong thing. The archived coefficient vector on which every kernel diagnostic rested is reproducible—the rebuilt cache returns it at correlation +1.0000—but it is a self-kernel estimated *jointly with* 41 exogenous channels, several of which are direct volatility measures. Conditioning on near-substitutes for lagged RV absorbs the target’s persistence and leaves a contrast, which is what produced the negative lobes and the inadmissible integral. The marginal self-kernel on the same rows has integrated mass +0.913 and no negative mass (Section 6.6). We regard this as the most transferable methodological point in the paper: a self-kernel estimated alongside volatility proxies is not a self-kernel, and the failure is silent.

Structural claims are not identified, and one of them is refuted. We do not fit a Hawkes process. Section 6.4 establishes a correspondence between our regression and a Hawkes intensity and reports quantities that would be structural under it. Testing them splits the correspondence in two.

The *shape* survives testing. The decay exponent, previously a hardcoded plotting constant, is estimated at 1.033 with a channel-clustered standard error of 0.036 and agrees with the Hawkes-in-finance literature; the power law beats an exponential on fit and on forecast; the exogenous operator is rank 4–6 of 41 channels.

The *sign structure* fails it, on the archived vector: eight of twelve lags negative, 10.3% of absolute mass pooled and up to 87.8% within a session, close to half of all exogenous cross-kernel cells negative, and an integrated kernel of -0.038 whose sign flips across sessions.

Nested probes on the rebuilt cache isolate the cause as the exogenous conditioning set (**[pending: tab:nested_probes]**): the marginal kernel is +0.913 with no negative mass,

adding the 41 exogenous ladders takes it to -0.139 , and the full archived set to -0.363 . The sign structure is a property of conditioning on volatility proxies, not of the target. We report $\hat{n} = 0.913$ as an analogue rather than an identified branching ratio, because the estimate is ridge-penalised on transformed data and because a 30-minute aggregated variance is not an event stream.

A rival benchmark is missing. A fractional-kernel forecaster sits in the same power-law family as our backbone (Section 6.8) and is absent from the battery. Given the paper’s one-benchmark discipline this is the single most defensible addition to the ladder sweep, and it is inexpensive.

14.3 Economic Significance

Every effect in this paper lives between QLIKE 0.126 and 0.134. The joint exogenous information effect is 0.00627, or 4.7% relative to the incumbent; the model-class effect is 0.00184, or 1.4%; the hyperparameter oracle ceiling is 0.00015, or 0.1%. We have not translated any of these into a quantity that a user of the forecast would act on, and we think the ordering *information > model class > hyperparameters* is the paper’s transportable content precisely because the absolute magnitudes are small and panel-specific.

Two considerations argue against over-reading the levels. First, the calibration result of Section 8.6 shows that QLIKE rewards attenuation on this panel, so an arm’s QLIKE advantage is not straightforwardly an advantage in forecasting realized variance. Second, all reported levels are out-of-sample in the walk-forward sense but in-sample with respect to the choice of preprocessing pipeline, feature dictionary and evaluation window, none of which were pre-registered.

14.4 Reproducibility

Model results in this version derive from a single frozen battery of 98 arms on one panel of $n = 218,934$ bars, with per-arm metrics archived alongside the manuscript. Superseded experiments—in particular the hyperparameter search of Section 8.4 and the variable- N subgroup tables—are retained in the repository history but contribute no number to this paper.

All inference in Section 11 is reproducible from the archived artefacts. The estimators live in `analysis/inference.py` (QLIKE, Newey–West long-run variance, Diebold–Mariano with the Harvey–Leybourne–Newbold correction, the Politis–Romano stationary bootstrap, the Hansen–Lunde–Nason MCS, Holm and Benjamini–Hochberg); they are validated in `analysis/test_inference.py` against `scipy` for the t distribution and `statsmodels` for the HAC standard error and both multiplicity procedures, plus an empirical size check of the DM test at the nominal level. The two drivers are `analysis/run_tile_inference.py` and `analysis/run_battery_bounds.py`; `analysis/make_tables.py` regenerates every inference table in this paper directly from their JSON output, so no statistic in Section 11 is transcribed by hand. Bootstrap replications are fixed at $B = 5,000$ under seed 20260802, and the block length is chosen automatically with reported sensitivity from half to four times its value. Environment: Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, statsmodels 0.14.6.

The frozen battery itself is not reproducible from this repository: it was produced on two HPC clusters and only its per-arm summary metrics were retained, which is precisely why Section 11.2 must bound the pairwise test rather than compute it. Persisting per-bar loss series should be treated as a requirement of any future battery run, not an optimisation.

15 Conclusion

This paper asked four questions of a single fixed panel of 218,934 thirty-minute S&P 500 realized-variance bars. It answers a fifth it did not ask, and that answer subsumes much of the rest: almost all of the modelling apparatus we built can be deleted without loss, and the effects that remain are smaller than the protocol and data-quality effects surrounding them.

What should the instrument be? A truncated power-law kernel with a clock organ. The fitted HAR coefficients imply a per-lag effective kernel that is approximately power-law on log-log axes, which recasts the lag ladder as a quadrature grid rather than a set of economically distinct horizons. The recasting is not decorative: it predicts that refining the grid pays, and the grid refines monotonically across six geometric bases, with base-2 beating the base-5 incumbent by 0.00082 at DM $t = -10.6$ and the conventional daily/weekly/monthly structure faring worse than either. It also predicts that a power law should beat an exponential, which it does by 0.011 in forecast loss and decisively in kernel shape. Because the shared benchmark uses the coarser grid, we carry an explicit 0.00082 of benchmark debt through every downstream increment rather than absorbing it.

The self-exciting reading that the power-law shape invites is tested rather than asserted, and it splits in two. The shape survives: the decay exponent, estimated across 41 exogenous channels where the log-log slope is invariant to channel scaling, is 1.033 with a channel-clustered standard error of 0.036, agreeing with the Hawkes-in-finance literature and rejecting both the 1.36 earlier drafts had carried as a plotting constant and the ≈ 1.6 that a rough-volatility correspondence implies. The sign structure appeared to fail, and explaining why produced the section’s most transferable result. The estimate this project had been reading conditions the self-kernel on 41 exogenous channels, several of which are direct volatility measures; those absorb the target’s persistence and leave an integrated kernel of -0.363 with 14% negative mass. The marginal self-kernel on identical rows has integrated mass 0.913, no negative mass, and monotone power-law decay at $\hat{\beta} = 1.506 \pm 0.035$ over ten lags—subcritical, close to unity, and squarely where the reflexivity literature places financial activity. A self-kernel estimated alongside volatility proxies is not a self-kernel, and the fit gives no signal that anything is wrong.

Where does the information live? Everywhere and nowhere in particular. Every exogenous bucket improves on the backbone with overwhelming individual significance, yet the buckets overlap so heavily that the sum of within-class gains is 2.08 times the joint gain. Within the least-squares class, 96% of explainable variance belongs to the backbone; the 359-column exogenous block adds a unique contribution that a single mechanism-targeted engineered feature exceeds by roughly two orders of magnitude per column—a ratio we report with its arithmetic and its caveats rather than as an effect size. The individual feature landscape repeats the pattern one level down: one dominant cheap feature, one partially redundant companion, a long tail of tiny effects, and a genuine null.

Does extraction require nonlinearity? Yes, but boundedly, and the bound is the interesting part. Tree ensembles beat the best penalized linear model on all nine feature sets, by between 0.00184 and 0.00358; the consistent sign is the evidence, and the pairwise difference remains untested. The pure nonlinearity premium on backbone features is 0.00215 measured against a correctly specified linear ladder, some 28% below the figure obtained against the incumbent. Tuning is a non-lever from every angle: causally tuned linear arms are worse than the untuned incumbent, and a test-set oracle over penalties moves it by at most 0.00015. What the trees exploit saturates at additive-plus-pairwise structure anchored to the session clock, and deeper capacity, deep sequence models, and mixture-of-experts routing all fail to improve on it—a ceiling that a quadratic self-exciting feedback predicts. The resulting ordering—information > model class > hyperparameters, by roughly a factor of three at each step—is the paper’s central practical law.

What algorithm should do the extracting? With features frozen, exactness and anchoring matter more than expressiveness. Rank-1 rolling least squares makes the every-bar OLS benchmark computable at scale; the online elastic-net homotopy extends the same exactness to ℓ_1 estimation; and the causal tuning protocol converts “tuning does not help” from an anecdote about one search into a claim with a defensible causal structure. The spectral k -nearest-neighbors proposal is instructive precisely because its ablations are decisive against it: the OLS anchor is load-bearing (worth up to 0.0132), large-neighborhood averaging is load-bearing (worth ≈ 0.0018), and the manifold embedding—the novel part—hurts on all eight matched configurations. A matched retrieval-metric experiment then locates the cause, and it is not the out-of-sample extension error we had blamed: a principal-component projection at the same dimension has no extension step and loses nearly as much, while a supervised coordinate of that dimension ties the full ambient view. What fails in this regime is unsupervised geometry, at any dimension and whether linear or nonlinear—and, relatedly, breadth in a distance, since dropping the exogenous block entirely improves retrieval by 0.01476 ($t = -15.5$). A fitted map with learned weights profits from many weak channels; a metric that weights every coordinate equally is buried by them. What survives is an unembedded, anchored analog search that beats the incumbent at QLIKE 0.13096; it also beats 25 of the 45 causally tuned parametric arms, while trailing the best arms on the informative buckets. In the dense-but-weak regime, nonparametric flexibility is a cost to be justified, not a default.

How much of the apparatus is load-bearing? Almost none of it. The 41×12 exogenous operator, its singular value decomposition, the session-conditional frames, the CP and Tucker factorisations, the rank-constrained priors and the bespoke projection penalty are together reproduced by contracting each channel’s rung ladder against a small number of fixed shapes and running plain ridge on the result. On the uncorrected panel that compression appeared to *beat* the full design—53 columns at 0.13280 against 0.13617 for plain ridge on all 504 features, itself worse than the 12-column backbone at 0.13571—and we built a draft around it. On the corrected panel the full design reaches 0.13146 and wins, our pre-registered falsification condition for the compressed one is met at -0.00126 ($t = -5.91$), and the compression is withdrawn (Section 13).

What replaces it is a tie. Sweeping rank on the lag axis under two independent orthonormal bases, the curve is flat from $r = 4$ upward, and the best compressed arm—135 columns—reaches 0.13117 against 0.13146 for all 504. That 0.00029 does not survive a widened HAC

bandwidth, a block bootstrap, an era-level test or a leave-one-era-out sweep, so we report the two as equivalent. A nearly fourfold difference in design width buys nothing measurable, which is this paper’s thesis applied to its own modelling section. The effective-degrees-of-freedom accounting says why: the elaborate penalty spends ≈ 54 degrees of freedom, which is $1 + 12 + 41$, the dimension of its own null space. The compressed design is an order of magnitude more robust in the one era where the exogenous block fails ($+0.00347$ against $+0.03321$). The generalisation we are willing to defend is that in this regime every *estimated* structure is a cost requiring evidence: eigenmaps, principal components, the operator’s own leading direction re-estimated or frozen, and rank-one factors with fitted loadings all lost to fixed alternatives, without exception.

What dominates all of it? Protocol and data quality—where they bind, by orders of magnitude; on average, by less than we first claimed. A fixed training window rather than a rolling one is worth 0.11 QLIKE. The imputation defect of Section 4—an availability indicator computed after a forward fill, which silently disabled three guards this codebase already contained for that exact pathology—put scaled features at 2,260 rolling interquartile ranges and cost a factor of fifty on its window, 6.37170 against 0.12391. Selecting a shape parameter against the reported loss rather than a validation tail is worth 0.0033. Against these, the model-class effects this literature argues about are 0.002 to 0.004.

We pre-registered the stronger form of this claim and it failed. Pooled over six designs, the data-version main effect is 0.00067 against a design main effect of 0.00311: averaged across twenty years and across designs that mostly do not touch the affected columns, data representation does *not* dominate. What dominates is the conditional statement—where a fill defect binds, it is worth more than every modelling choice combined, and nothing in the fit tells you whether it binds. That is the transferable finding, and it is narrower and more useful than the one we set out to make: a team that checks its availability bookkeeping against its raw feeds will not lose fifty-fold to a dead channel, and no amount of care about model class substitutes for that check.

And is any of it stable? Not as reported. The exogenous block beats the backbone in seven consecutive eras and fails once, so a twenty-year aggregate supports opposite conclusions depending on where the window ends, and 27 bars out of 27,376 carry 47% of the failure. We regard single full-sample aggregates over multi-decade panels as insufficient reporting, in this paper and in the literature it follows.

On the evaluation itself. Two findings concern the measuring apparatus rather than the models, and we think they travel beyond this panel. QLIKE and raw-variance MSE rank our 45-arm battery almost independently (rank correlation $+0.15$, $p = 0.32$, where agreement would give a negative value), and 14 of the 42 arms that improve on the incumbent under QLIKE are worse than it under MSE. Separately, QLIKE systematically rewards attenuation: the rank correlation between QLIKE and the Mincer–Zarnowitz slope is $+0.74$ ($p = 5.8 \times 10^{-9}$), so the best-scoring arms are the most miscalibrated, with the second-best arm in the battery carrying its worst calibration. Neither observation invalidates a strictly consistent scoring rule, but both change what a QLIKE ranking should be taken to mean.

On the cost of testing what we claimed. Section 11 subjects the paper’s inherited claims to per-bar inference for the first time. Four survive Holm-adjusted testing, two are refined, and three do not survive—trees as residual readers, Fourier clock modulation over session gates, and rank versus divide were all directionally correct point estimates that had been promoted to results. A fourth, a singleton model confidence set over feature sets, turned out never to have been computed at all. We report the casualty list because it may be the most transferable finding here: on a 2,189-bar tile a model confidence set retains 123 of 161 arms, so a tile leaderboard is not a ranking, and a difference read off one is real only if it replicates at scale.

What replicated. The strongest evidence for the paper’s central claims is no longer any single table but Section 12: on a second basis, with different machinery, under pre-registration with recorded expectations, the same laws re-derive — dense-generous-shrunk beats every selective alternative it met (sixteen more retired arms), the exploitable content is fast in horizon and rewards per-bar freshness in the first moment while the second moment obeys neither law, and every estimated geometry without supervision lost to a frozen one, including the retrieval corrections whose panel-specific success Section 9.3.4 could not confirm. The replication also sharpened the era story: regime change arrived there not as decay of predictability but as collapse of dictionary complexity, a form of nonstationarity that headline accuracy metrics are structurally unable to see.

Open items. We flag the incomplete edges of the evidence explicitly, and Section 14 treats them at length. (i) The bucket battery has not been re-run on the base-2 backbone; until it is, every increment carries 0.00082 of benchmark debt and the smallest effects are not claimed. (ii) The model-class gap is now tested pairwise, by a bound that assumes nothing about the dependence between arms and resolves all nine matched comparisons after Holm adjustment; the headline pair is the weakest at $|t| \geq 2.19$, and a model confidence set over the 45 arms remains unavailable because their per-bar losses were not retained. (iii) The kernel estimate is reproducible and, once marginal rather than partial, admissible: $\hat{n} = 0.913$ with no negative mass. We report it as an endogeneity analogue rather than an identified branching ratio, because it is ridge-penalised on transformed data and because a 30-minute aggregated variance is not the event stream a point-process model describes. Two further limitations were settled along the way: the 5/95 target winsorization costs roughly 0.024 QLIKE and should be expected to disappear from the pipeline, and the rolling diurnal divisor used alone reaches QLIKE 0.558 against 1.320 for an unconditional mean, so it closes about 71% of the distance to a fitted model before any model is fitted. (iv) A fractional-kernel arm sits in the same power-law family as our backbone and is absent from the battery; it is the most defensible missing benchmark. (v) The winning spectral- k NN arm is the minimum of 32 cells selected on the evaluation panel, and survives on one bucket while sibling buckets failed as singular; the failure needs diagnosis before the bucket ranking can be trusted, and the replication panel’s pre-registered three-view test declined to confirm the family (Section 9.3.4). (vi) The attribution results point outward: the highest-value next investment is not a larger model but mechanism-targeted data—auction imbalance, order flow, and dealer positioning aligned to the session-transition regime where the nonlinear

signal concentrates.

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